



Universities Press

REFERENCE

THE STORY OF OUR FOOD



K T ACHAYA



THE STORY OF OUR FOOD

[For more books from India on the iTunes store click here](#)

[*OceanofPDF.com*](#)

THE STORY OF OUR FOOD

K T ACHAYA



Universities Press

OceanofPDF.com

Universities Press (india) Private Limited

Registered Office

3-6-747/1/A & 3-6-754/1, Himayatnagar, Hyderabad 500 029 (A.P.), INDIA

e-mail: info@universitiespress.com

Distributed by

Orient Blackswan Private Limited

Registered Office

3-6-752 Himayatnagar, Hyderabad 500 029 (A.P.), INDIA

Other Offices

Bangalore, Bhopal, Bhubaneshwar, Chennai,
Ernakulam, Guwahati, Hyderabad, Jaipur, Kolkata,
Lucknow, Mumbai, New Delhi, Noida, Patna

© Universities Press Private Limited 2000

First published 2000

eISBN 978 81 7371 835 9

e-edition: First Published 2012

All rights reserved. No part of this publication may be reproduced, distributed, or transmitted in any form or by any means, including photocopying, recording, or other electronic or mechanical methods, without the prior written permission of the publisher, except in the case of brief quotations embodied in critical reviews and certain other noncommercial uses permitted by copyright law. For permission requests write to the publisher.

OceanofPDF.com

CONTENTS

[List of Illustrations](#)

[Preface](#)

[1: Our Early Ancestors](#)

[2: The Harappan Meal](#)

[3: How Weeds became Foodgrains](#)

[4: The Aryan Way of Life](#)

[5: Theory and Practice](#)

[6: Through Other Eyes](#)

[7: The Muslim Contribution](#)

[8: Many Regions, Many Foods](#)

[9: Europe Views the Indian Scene](#)

[10: The Taste of the New World](#)

[11: More Exotic Delights](#)

[12: Bibliography](#)

[13: Glossary of Indian and Non-English Words](#)

[OceanofPDF.com](#)

ILLUSTRATIONS

Fig 1: Bhimbetka Cave Drawings

Fig 2: Some Harappan Towns

Fig 3: Developing New Sugarcanes

Fig 4: Possible Aryan Migration Routes

Fig 5: Flora Depicted in Buddhist Sculpture

Fig 6: Two Designs of the Ghani

Fig 7: The Rice Plant Astonished the Greeks

Fig 8: Rohu. Indian Fish were Praised by Babar

Fig 9: The Emergence of Indian Scripts

Fig 10: Pepper Lured the Europeans to India

Fig 11: How American Food Plants Reached the East

Fig 12: The Groundnut Develops Below the Ground

OceanofPDF.com

PREFACE

Behind any of the foods that we eat every day, lies history and geography, botany and genetics, processing technology and high romance.

The ingredients of the food themselves have come to us from so many parts of the world. Some of them have travelled across continents and across oceans. For thousands of years, food plants have made their way to India to be adopted, adapted, nurtured and cherished here. Visitors to our country from ancient times, whether Greek or Chinese, Arab or European, have expressed their astonishment at the wide variety of food that they found in India, and its sheer profusion.

We still cook our food in ways developed by our predecessors. The ethos of food preparation, and the attention to cleanliness and purity are, in large measure, an Aryan inheritance that goes back three thousand years. For the last one thousand years, another culture, that of the Muslims, entered our country and gave new and richer dimensions to Indian cooking.

Frying pans, *kadhais* and *tandoori* ovens go back thousands of years. So do mechanical appliances for pounding, grinding and the pressing of sugarcane juice and vegetable oils.

In the last five hundred years, from Mexico and South America, came food materials that we now take for granted, like the groundnut and cashewnut, potato and tomato, and yes, even the chilli! All of which today no Indian household can do without.

This little book has been written to help you share some of the excitement that lies behind the food of India, and the magnificent culinary legacy we have inherited.

Bangalore

Dr. K. T. Achaya

May 2000

OceanofPDF.com

1 OUR EARLY ANCESTORS

Man in India

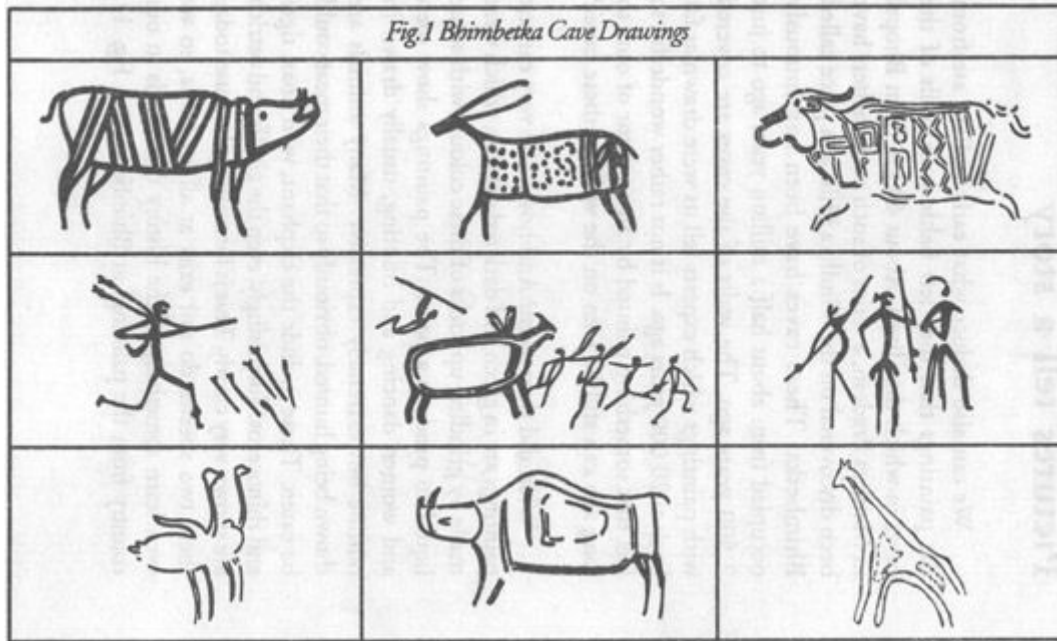
It is obvious that monkeys and men resemble each other. Millions of years ago, a common ancestor started on the road to becoming man, walking upright, and developing a larger brain and logical thought. About 2 million years ago man emerged in his modern shape and form, called *Homo sapiens*. Possibly this occurred in Africa, followed by migration in various directions. We learn all this when fossil bones of these early ancestors of man are discovered.

They also left various implements which come to light at times. The large apes, from whom man is descended, were and are vegetarians, but these tools tell us that early man soon became a hunter of animals. The earliest tools were huge, blunt stones called chopping tools which were in use about half a million years ago. In the next stage of his development, man started collecting food for his use. The tools then changed to pointed oval stones which were employed as hand-axes and digging tools, and as knives and scrapers of flesh from skins. In the next stage man stopped foraging for food, and started to grow his own. Small, sharp stone chips called microliths, fitted to wooden handles, were used for a variety of purposes. They were used to cut down crops, to chop up meat and vegetables, to tip arrows shot from bows, and so on. Animal bones are found in every archaeological site that has ever been excavated in India, but later, as man progressed, meat-eating was combined with the use of agricultural crops. The earliest Indian culture was the Indus Valley civilization, the cities of which flourished on the banks of many rivers in Punjab, Gujarat and Rajasthan from 2500 BC for about a thousand years. The bones found at these sites showed that beef, mutton, turtles, small Indian alligators (gharials), and river and sea fish were all eaten in addition to foods like wheat and barley. While the bones of the domestic fowl have been found at these sites, some sort of taboo against eating it has persisted right down the centuries. But we do know that every chicken in the world today is descended from the Indian jungle fowl, which was their original ancestor.

Pictures tell a story

We can also deduce what early Indians ate from the paintings that they left behind on walls of the caves in which they lived. About 40 km from Bhopal in Madhya Pradesh, a series of such rock shelters have been discovered on the Vindhya Hills at a place called Bhimbetka. These caves have been continuously occupied from about half a million years ago to just 2 500 years ago. The walls of the caves are covered with paintings which experts tell us were drawn as far back as 20 000 years ago. Is it not rather wonderful to feel that something painted by an ancestor of ours so long ago can still be seen on the walls of these caves?

What did they paint? And how? The very earliest paintings are in green and dark red colours, which were made by grinding up rocks of these colours with some liquid to produce a paint. The paintings show men and women dancing and hunting, usually drawn in outline but extremely expressive. Many animals are shown being hunted, obviously so that their meat could be eaten. These include the elephant, wild boar, tiger and rhinoceros. Surprisingly, even the giraffe and ostrich are shown very clearly. This is interesting, because today these two species do not exist at all in India, so we even learn something of the history of animals in our country from the paintings at Bhimbetka (see Fig. 1). Hunting scenes, painted slightly later, show hunters using barbed spears, pointed sticks, bows and arrows with a variety of animals as the victims. Meat of many kinds was thus eaten, probably supplemented with wild fruit, vegetables and greens which could be collected from neighbouring forests. In fact, even today, many of our tribal people live in exactly the same way. They catch all kinds of animals, even small mice, lizards, crocodiles and snakes to supplement the many edible fruits, leaves and roots which they collect from forest trees and plants. Both the tools left by our ancestors, and the pictures that they painted, tell us quite a lot about the kind of food which was eaten long ago.



Words, too, are ancient

We have one more source from which to trace the foods of our forefathers. All languages evolve and grow. Some of the words which we use today are thousands of years old, and when such a word is located, we can be sure that the object existed thousands of years ago.

The early inhabitants of India are loosely spoken of as the Mundas. They were spread out all over the country and spoke a number of languages which, however, are now called the Munda language for convenience. Later, the people of the Indus Valley civilization, which we have just read about, developed a language of their own. We know how this language was written from seals that these people left behind and which have been excavated. However, though many attempts have been made to decipher the language of the Indus Valley people, so far no one has succeeded in doing so. The next major language of India was the one brought in by the Aryans who came in waves into the country from about 2 000 BC onwards. This language became Sanskrit: it is, therefore, the oldest known language of India, and we have works composed in it from about 1 500 BC. In the south of India, the oldest spoken language was Tamil, but the literature that has survived is only from about 200 BC. Let us look now at some of the words concerned with food which occur in these two languages.

The Aryans referred to the Mundas by various uncharitable names, one of which was "Nisada". This comes from two words, "nisa" meaning turmeric and "ad" meaning

to eat, so we do know that turmeric or haladi existed long before the Aryans came to India. One tribe of the Mundas was described as "Svanin", which means dog-keepers, and another as "Punjistha", which means fowlers, so both the dog and the fowl had been domesticated by that time. The turmeric itself, which we now call haladi in Hindi, was called *haridra* in Sanskrit, and is itself an old Munda word adopted into Sanskrit. So is the word for the lime, which was first *numbaka*, then *nimbaka*, and now *nimbu*, so here again we have a very old food. Both the betel leaf (*thambula*) and the areca nut (*guvaka*) were Munda words, so betel-chewing itself must be very old in our country. The tamarind was called *chinchapala* in Sanskrit: this is also a very old Munda word, and so is *injivera* for ginger. Brinjal was *vrntaka*, a Munda word. The very word *chaval* for rice is supposed to come from *chom-la*, which is itself derived from the Munda word *jom* meaning 'to eat'. In fact, the names for rice are all very interesting. The Sanskrit word for rice is *vrihi* or *varisi*; from *varisi* came the Tamil word *arisi* which went along with rice itself from south India to Europe, and gave rise to the word rice. The English word 'curry' is also nothing but the Tamil word *kari* which means a dish cooked with pepper, so both curry and rice are good old Tamil, Indian words. You will probably be surprised to know that this is also true for sugar. Sugar is an Indian food first developed in this country, and the original name was *shakkara*, which became today's sugar. And of course mango is Indian, the Sanskrit words being *amra*, *oma* or simply *ma*, from which came the Tamil *maanga* and the English mango. The Munda name for the jackfruit, which was *panasa*, also entered the Sanskrit language.

Sometimes, a word coined for a specific purpose later became a common noun. Let us look at some examples of this kind. The Sanskrit word for sesame or gingelly is *tila*. The oil from the sesame seed was called *taila*, and gradually this became a general word for all oils in north India. Similarly, in Tamil, the word *nai* originally meant coconut oil, and even before that it meant the coconut itself. But today *ennai* is used in Tamil for all vegetable oils. Ghee is now called *nai*, and butter is called *vennai*. Two other examples: honey in Sanskrit was originally *madhu*, but anything sweet is now called by the same term. Similarly, the Tamil word for tamarind, an old article of food, is *pulr*, *puli* in Tamil is now applied to anything that is sour.

Among all the foods that we eat, there are at least some things which have been handed down to us from very long ago. The story of our food does go back a long, long way in history.

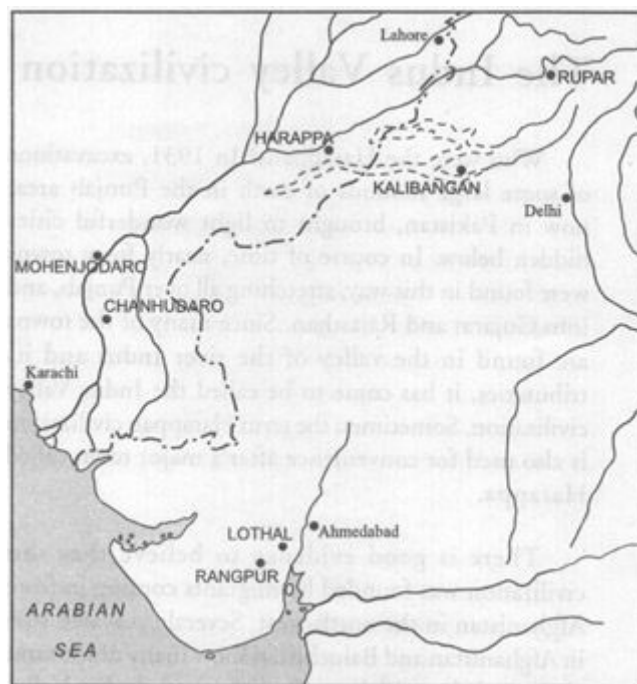
2 THE HARAPPAN MEAL

The Indus Valley civilization

Who were the Harappans? In 1931, excavations of some large mounds of earth in the Punjab area, now in Pakistan, brought to light wonderful cities hidden below. In course of time, nearly forty towns were found in this way, stretching all over Punjab, and into Gujarat and Rajasthan. Since many of the towns are found in the valley of the river Indus and its tributaries, it has come to be called the Indus Valley civilization. Sometimes, the term Harappan civilization is also used for convenience after a major town called Harappa.

There is good evidence to believe that this civilization was founded by migrants coming in from Afghanistan in the north-west. Several excavated sites in Afghanistan and Baluchistan show many of the same characteristics which marked the later Indus Valley civilization. Some of these are a kind of pictographic writing, domestication of the goat, sheep and buffaloes, the use of barley, wheat and cotton and the storage of foodgrains.

Fig. 2 Some Harappan Towns



Among the bigger towns were Mohenjodaro, Harappa and Lothal. The first two were situated on rivers, and the last was a sea-port. (See Fig. 2). The towns were laid out in a surprisingly orderly, modern fashion, with roads running at right-angles to each other. Every home had a bathroom with excellent drainage, and there were large public baths as well. Enormous granaries were found for storage of food-grains. At Mohenjodaro, this took the shape of 27 storage blocks arranged crosswise to ensure good air circulation, which is very important for long-term food-grain storage. The whole granary was built on a platform made of bricks, and grain was unloaded from carts standing below and transferred to the storage blocks. Thus, food-grains were available all year round for feeding the people—a remarkable achievement for a period so long ago. In fact, the Indus Valley civilization lasted almost exactly a thousand years, from 2 500 to 1 500 BC or so.

At Lothal, the warehouse overlooked a huge dock, lying near the sea. The size of the warehouse was enormous, 51 by 45 metres each way, and the actual storage blocks here were 12 cubical platforms. On each of these probably stood a wooden superstructure. Melted material was found in the ruins, suggesting that a huge fire must have destroyed all the wooden structures, leaving only the brick platform.

Near the storage granaries at Mohenjodaro were found a large number of circular grinding platforms made of brick, laid edgewise. In the centre of each was a hole in which the grains were pounded, using wooden pestles which have, of course, disappeared. Grains of burnt wheat and husks of barley were actually found still lying in the crevices of these platforms. Just imagine, these were the same grains that were actually eaten by people nearly 4 000 years ago.

Foods of the Harappans

For grinding grain in India we still use large circular grinding stones made of granite, in which the top piece revolves above the lower one. Exactly the same kind of querns were found at Mohenjodaro and Harappa, showing how old are our food practices. Both at Mohenjodaro and at Lothal were found tandoori ovens used for baking, exactly as it is done today in the Punjab area. And at a site called Kalibangan, excavations brought to light a ploughed field ready for sowing. It has been called the oldest ploughed field in the world, and it is anybody's guess why it was got ready and then abandoned without being sown. The type of criss-cross sowing at right-angles to each other is exactly what is done in that area even today, where short plants like the mustard are grown in one line of furrows, and tall plants like bajra are grown at right-angles so that their shadows do not fall on the lower plants.

What were the foods that were in use in the huge Indus Valley area? The bones of many animals have been found, like cattle, sheep, turtles, tortoises, gharials, and river and sea fish. The bones of fowls have also been found. The beautiful seals of the Indus Valley, which were probably used to stamp traded goods, carry representations of hump-backed zebu cattle and of the water buffalo. It is, therefore, likely that all these were eaten by the dwellers then.

Among the crops grown, wheat and barley have already been mentioned, and they seem to have been common foods all over the area of the Indus Valley culture. Rice was grown only in the more southerly towns, such as Lothal and Rangpur in Gujarat. At certain southerly sites, ragi and Italian millet (called *kangni* in Hindi and *thennai* in Tamil) have been discovered, as well as crops like *jowar* (a millet). A large number of *dhals* with which we are now familiar have been excavated at the Indus Valley sites. These include *masoor dhal*, *urad dhal*, *mung dhal* horse gram and garden peas (*mattar*). All these are foods that we still eat and enjoy in India because our ancestors of long ago had the wisdom to choose and develop them.

What about cooking oils? Of course, the oils found in the meat of the animals that were killed must have been used for cooking in the Indus Valley, as they were everywhere else in the world. These included tallow from mutton and beef, lard from pork, and fish oil. It is also likely that the ghee from milk was used. However, even at this early stage, vegetable oils were being used in the Harappan cities. A lump of charred sesame seeds (*til* or gingelly) was found in the archaeological excavations of Harappa. At other Indus Valley city sites, mustard seeds and linseed have also been found. It is likely that the oils from these seeds were known too. Both sesame oil and mustard oil are important cooking oils in India even today.

What about fruit? There is plenty of evidence that many fruits were available then to the dwellers in the Indus Valley cities. Both lemons and melons were known. Dates, pomegranates and probably bananas were all eaten. It is very likely that the coconut was known, though perhaps it was not very abundant in regions so far north. Of course there must have been many other vegetables, berries, and green leaves that were also eaten; being perishable, they cannot now be traced or detected in excavations.

Trade from the Indus Valley

There is proof of extensive trade both by land and by sea from the Indus Valley cities with the great civilizations of the time in the Middle East, for example, the Sumerian civilization in the Euphrates Valley in Iraq. These two flourishing cultures exchanged goods with each other. In a house in Lothal were found certain objects which could only have come from Sumer. Similarly, Indus Valley seals, and Indian wood and ceramic beads, have been found in Sumerian excavations. The products that were exported from India were cotton, gems, pigments, gold and copper; what came back seems to have been mainly silver.

These trade contacts are of interest to us because they provided opportunities for foods, seeds and plants to be exchanged. The sesame plant and seed had their origin in India, and the seeds are mentioned in clay tablets found in Sumeria under the name *se-gts-i*, from which the word sesame was derived in Europe. Similarly, the linseed is called *i-gis*, and it could quite possibly have also travelled from India to Sumeria.

The end of a culture

Excavations in many Indus Valley towns show evidence of frequent flooding when the rivers on which they stood overflowed their banks. Also, the rivers in the Punjab area are well-known to change their courses as a result of upheavals below the ground. The fabled Saraswati river of the past does not exist any more, except as a dry river bed. In addition to this, the rivers began to silt up as a result of soil in huge quantities being carried down from the mountains; in consequence, the Rann of Kutch, into which the Indus Valley rivers flow, turned into a swampy marsh. Further, when rivers begin to silt, the soil around becomes filled with salt, and plants will not grow properly. It is also possible that extensive deforestation lowered the rainfall in the area. All these circumstances are believed to have brought to an end the great civilization which had lasted for about a thousand years.

There was one more factor. From about 1 700 BC, it is surmised that waves of a new people, the so-called Aryans, began to pour into India from their original homes in the steppe regions of South Russia. These same people also migrated to Europe and their descendants are the Europeans of today. The Aryans were fierce and warlike, and ruthlessly destroyed or pushed back the people whom they met in India. By about 1 500 BC, the Indus Valley civilization had vanished. Yet, some of the cultural patterns that were followed then have come down to us. The foods that were used are certainly among these.

OceanofPDF.com

3 HOW WEEDS BECAME FOODGRAINS

The great Russian botanist, Vavilow, about seventy years ago, identified what he called "centres of plant origin" in which the "evolution of plants was directed by the will of man." There were nearly a dozen of these centres all over the world where plants gradually took their place as foods for human beings. Of particular interest to India was the so-called Fertile Crescent of the Middle East, stretching from Israel to Iraq, which was an early centre of agriculture and of plant evolution.

What actually happened was this. Man would pick up promising weeds growing wild which carried grains. By choosing plants with abundant or plump grains, a process called selection, the quality of the grains and the yield of the plant in the next crop were both improved. Sometimes, nature itself would take a hand in the process; a wild weed would cross by chance with a cultivated species to produce offspring of a quality superior to both. Let us look at some famous examples where such things actually happened.

Wheat

The Wild grass ancestor of modern wheat was called *einkorn* because it had only one grain on each spikelet, and *ein* in German means one. From this wild form, a cultivated *einkorn* was grown, and like the wild variety was a diploid with seven chromosomes. Now nature gave a helping hand. A totally useless wild grass, called *Aegilops squarrosa*, crossed with *einkorn* to give tetraploid plants with 14 chromosomes. In fact, seven different tetraploid weeds were produced by this marriage, and some of them are still grown in India to give what are called hard wheats. These are excellent for making *semiya*, *pheni* and vermicelli. The third and last stage of the evolution of wheat was the crossing of some of the tetraploids with another wild grass, called goat grass or *Triticum tauschii*. This marriage produced wheats with 21 chromosomes, and is believed to have happened somewhere near the southern end of the Caspian Sea where this goat grass grows. It produced the excellent bread wheats of today, and three of them have been grown in India for thousands of years. One of these is the common wheat, *Triticum aestivum*, which is

used to make bread and *rotis* and is grown all over the world. The second is called shot wheat or *T. sphaerococcum*, a dwarf variety grown in North India. The third is club wheat or *T. compactum*, which was popular in India till about a century ago.

In other words, three wild grasses –*einkorn*, *squarrosa*, and goat grass –have pooled their genetic resources to create the wonderful foodgrain wheat, from which *chappatis* and *rotis*, *tandoori* and *naan*, have been made in India right from the days of the Indus Valley civilization.

Barley

Barley grains are also very old, having been found at Mohenjodaro. Barley grows in two forms. In one of these there are two rows of grain, and in another form, six rows. There are wild barleys also with two rows and six rows of grain, so it would be natural to think that these were the respective ancestors of the two cultivated types. This easy guess turns out to be the wrong one. Careful research in laboratories has established without doubt that it is the wild two-row form that gave rise in the past both to the cultivated two-row form, and to the cultivated six-row form. The wild six-row form arose much later when the six-row cultivated form crossed with the wild two-row barley.

The barley found in the Indus Valley cities of Mohenjodaro, Harappa and Chanhudaro, is the small six-row variety which is botanically called *Hordeum vulgare* var. *hexastichum*. This may seem very complicated, but really means common (Latin: *vulgare*) barley (Latin: *Hordeum*) with six rows (*hexastichum*). Barley is mentioned repeatedly in very early Sanskrit writings as *yava*, and was the chief article of food in those times. Today, it is not such a popular foodgrain, and rice and wheat have displaced it.

Ragi

Botanically, ragi is *Eleusine coracana*. It was born in Uganda in East Africa. How do we know this? For several botanical reasons, such as the existence of its wild ancestors, the long mention of the grain in tradition, and the fact that ragi figures in old religious ceremonies in those areas. Ragi is a tetraploid, and so is the African wild plant called *E. africanay* which gave rise to it. But the Indian wild plant, which is called *E. indica*, is a diploid which does not cross with ragi at all. Now what does this mean? Only this, that the Indian wild weed could *not* have given rise to the ragi plant in India. This plant must, therefore, have come to India from East Africa some time in the past. When did this happen? Ragi has been found in an Indian excavation which

is dated at 1800 BC, and at several other archaeological sites in central India of a slightly later date. We must, therefore, infer that some unknown benefactor brought this foodgrain from Africa to India in about say 2 000 BC. It also seems possible from archaeological dating that two other food-grains, jowar and bajra, also came at the same time to India from the same area in East Africa, where they were originally evolved. You see, therefore, how no country ever stands really alone; certainly our food has come to us from unexpected places.

Rice

It comes as a surprise to know that there are 100 000 varieties of rice in India alone, and twice that number in the whole world. Its evolution is another fascinating story. First of all, a common perennial wild grass, one that grows for several years, gave rise to a wild annual type, which comes up year after year from seed. This wild rice, called *Oryza nivara*, grows even now in standing water in small ponds and at the edges of lakes in several parts of India. Somewhere in the river deltas of the Indus, or the Ganges, or the Mahanadi, this wild annual grass was picked up by man for its tasty foodgrains, and transformed by selection into the cultivated annual rice plant which is called *Oryza sativa*.

But this is not the end of the story. As *sativa* was carried by migrating people all over the East, three races with separate characteristics arose as a result of different eating habits. These are termed ecogeographic races. In India, where we eat with our hands, the rice called *indica* was evolved which, on cooking, gave shining grains that are white and fluffy. In Japan, where chopsticks are used, a sticky type of rice was more convenient, and a variety called *japonica* was evolved. In other places in South-East Asia where noodles are the popular form in which rice is eaten, large grains of a semi-sticky type called *javanica* were developed.

Rice is the preferred staple food of south India, whereas wheat is today preferred in the north. However, this does not mean that rice evolved in south India. In fact, the archaeological discoveries suggest that rice was developed in the north and the east and came into south India at a somewhat later date. Before the arrival of rice, one of the staple grains appears to have been ragi, and there were also a number of other less common grains like *thennai* and *varagu* which have gradually dwindled in importance.

Dhals

Three pulses are mentioned in very old Sanskrit literature. These are the three Ms, namely *masha dhal* (*urad* or black gram), *mudga* (*mung* or green gram), and *masura* (*masoor* or red gram). The first two of these originated in India itself from their wild ancestors, while the third one was developed in the Fertile Crescent but came into India before 2 000 BC.

The common *tuvar* or *toor dhal*, which is botanically called *Cajanus cajan*, was for a long time a bit of a puzzle. No wild forms have been found in India, and it was thought that *tuvar* was an African native. Then came an exciting discovery: a wild plant of a different species, called *Atylosia*, which is quite common in south India, was found to be the immediate ancestor of *tuvar dhal*. It had the same chromosome number, it crossed easily with *tuvar* itself, and the crosses were totally fertile. These were positive proofs that *tuvar dhal* arose from *Atylosia*. Even the Sanskrit name *tuvarika* is derived from the south Indian word *tuvarai* even now in use.

The *moth* bean or *matki* is not of great importance but is of interest for another reason. Both its wild and cultivated forms are found in India, and it is also mentioned in about 1 000 BC in Sanskrit literature. Yet this same *dhal* has also been found in South America in excavations dating from about 3 000 BC. How did India and South America, two countries so far apart, come to share a common domestic foodgrain so long ago?

Oilseeds

You will remember that a charred lump of sesame seeds was found in the excavations of Mohenjodaro dated at about 2 000 BC. For a long time, there was much speculation among botanists whether the sesame plant was indigenous to India or not. Today most doubts have been set at rest, and the plant is definitely known to have been developed in India. This occurred from a wild ancestor which even now grows quite commonly in various parts of the country. The botanical name of sesame, which is *Sesamum indicum*, was given to it by the great Swedish botanist, Linnaeus, about 200 years ago. It has turned out to be an accurate choice. From India, the sesame went a very long time ago to various parts of the Middle East, being referred to as *se-gis-i* in the clay tablets of the Sumerian civilization of about 3 000 BC.

Also very old is the little, black mustard seed which we use in our cooking every day. Charred seeds have been found in the Indus Valley city of Chanhudaro, dated to about 1 500 BC. This is a pungent variety called *rai*, and there are many references to it in Sanskrit literature right down the years. Botanically, *rai* is *Brassica juncea*, and it is a diploid with 36 chromosomes. This form originated by the straightforward fusion of the nuclei of two other diploids, one with 16 and the other with 20 chromosomes.

What is not very clear is whether this happened in Central Asia or in India, but wherever it was, it was a very long time ago. Apart from *rai*, we also have in India two other seeds, called *sarson* and *toria*. Both these belong to a related variety called *Brassica campestris*, which has 20 chromosomes and is, in fact, one of the two ancestors of *rai*.

The whole family *Brassica* is an extremely diverse one. Cabbage and cauliflower, knol-khol and turnip, all belong to it. In Punjab, a very popular pungent dish is *sarson-ka-saag*, made from the leaves both of *rai* and *sarson*.

Coconut

We have here a really old plant, which is believed to have been fully evolved even 15 million years ago, because fossilized coconuts have been found in excavations of these very old strata. All the evidence points definitely to the origin of the coconut somewhere near Papua New Guinea. The nut grows near the sea, and it can be carried by sea-currents to far-flung regions all over the southern hemisphere. It has been proved in modern experiments that even after months of floating in sea water, as soon as the coconut is washed up on a seashore it will begin to sprout, and a tree will eventually come up. It is likely that it came to India in this way without the agency of man. In fact, the Tamil name *tenga* means "sweet fruit from the south", and of course the direction from which the nut came to south India was definitely the south.

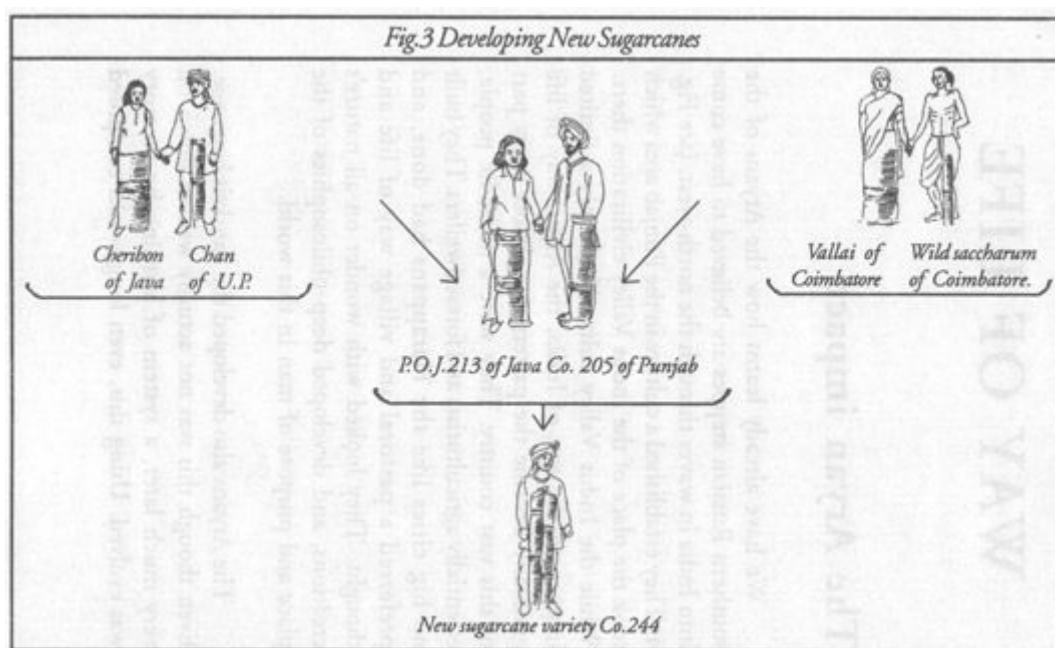
The coconut was long believed to be an Indian plant, since the Sanskrit name *narikela* (now *nariyal* in Hindi) has followed it wherever the nut was carried. We now know that the name *narikela* is a pre-Sanskrit Munda word, itself derived from two Southeast Asian words. One of these is *niyor* meaning coconut oil, and the other is *kolai* meaning nut. And when the word *niyor* came to south India it was transformed into the word *nai*, and later into *ennai* which is now used in Tamil to mean any oil. The evidence of language makes it likely that coconut oil was the first vegetable oil used in south India for cooking.

Fruits

The banana has at least one parent that is truly Indian. What happened in history was that a wild species of small fruits called *Musa acuminata*, which carries the genome labelled A, spontaneously gave rise to a diploid AA and a triploid AAA, both with small fruits. When these species came to India, they crossed with a plant carrying the B genome, called *M. balbisiana*, to give diploids like AB and triploids like AAB and ABB which carry large fruits. Today, there are two diploid and nine triploid

banana varieties grown in India. Their fruits vary all the way from small, plump, sweet varieties to big, tapering, green forms, and again to the really long, reddish fruit that makes such lovely banana chips.

The sugarcane has a somewhat similar history. Papua New Guinea, was again the home of the wild *Saccharum robustum*, from which sprang the thick sugarcanes called *S. officinarum*. This species migrated to India and was grown here for many centuries. Some sixty years ago, man took a hand in the business. At the Sugarcane Research Station, Coimbatore, the sugarcane was crossed with a wild grass called *S. spontaneum* to greatly improve the thickness, sugar content, and resistance to disease and adverse climate. In turn, these progeny were crossed back and forth with others to give a variety of canes. These were called by various numbers like Co.244, Co.290 and Co.301 to denote that they were evolved at Coimbatore. Colourful illustrations were used to publicize these varieties among sugarcane cultivators and the public. Today, these Indian-bred sugarcanes are the mainstay of the sugar industry all over the world, except where the sugarbeet is preferred as a sugar source (see Fig. 3).



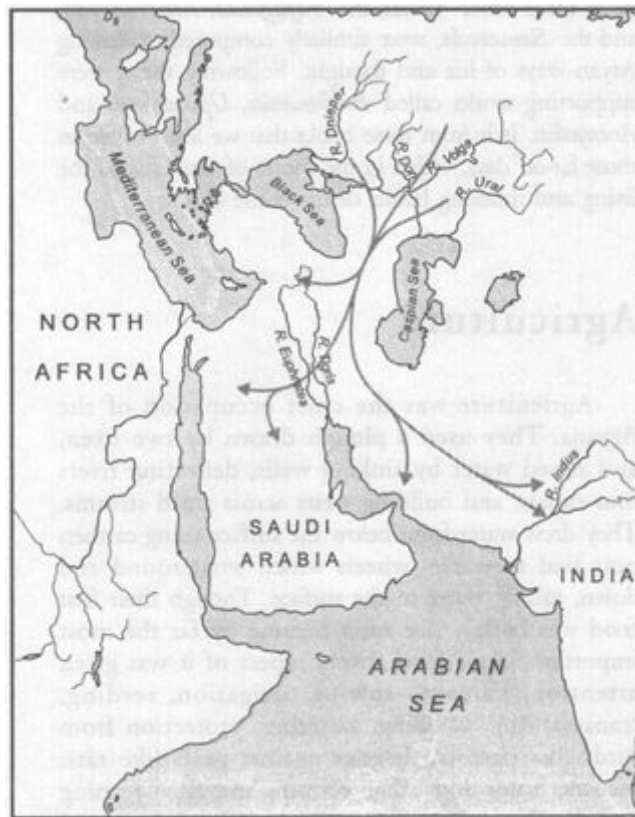
4 THE ARYAN WAY OF LIFE

The Aryan impact

We have already learnt how the Aryans of the southern Russian steppes are believed to have come into India in waves through the north-west, (see Fig. 4) They established a culture in the Punjab area which took the place of the Indus Valley civilization there. While the Indus Valley culture had only a limited impact on the rest of India, the Aryan way of life gradually became the pattern for almost every part of this vast country. They were a nomadic people, essentially agriculturists and forest dwellers. They built no big cities like the Harappans had done, and preferred a pastoral and village way of life and thought. They looked with wonder on all nature's creations, and developed deep philosophies of the place and purpose of man in this world.

The Aryans also developed the Sanskrit language. Even though this was not actually written down till very much later, a system of learning by memory was evolved. Using this, even long works composed in Sanskrit were passed down by memory from person to person from those far-off days down to the present. The first work composed was the *Rigveda*, which has been placed at about 1 500 BC. In the next few hundred years three other *Vedas*, the *Yajurveda*, *Atharvaveda*, and the *Samaveda*, were similarly composed expressing Aryan ways of life and thought. Following these, were supporting works called *Brahmanas*, *Upartishads* and *Aranyakas*. It is from these books that we learn of life in those far-off days, which in the course of time shaped the living and thinking habits of the entire country.

Fig. 4 Possible Aryan Migration Routes



Agriculture

Agriculture was the chief occupation of the Aryans. They used a plough drawn by two oxen, and raised water by sinking wells, deflecting rivers into canals, and building weirs across small streams. They drew water from below the surface using earthen pots tied to water -wheels which went round and down, raising water to the surface. Though their first food was barley, rice soon became by far the most important staple food. Every aspect of it was given attention, namely—sowing, irrigation, seeding, transplanting, weeding, watering, protection from birds like parrots, defence against pests like rats, locusts, borer and other vermin, and final reaping and threshing. The Aryans quickly learnt the use of fertilizers like cowdung, and later of oilcakes. They even practised the modern technique of silaging green leafy matter and straw, burying it underground without access to air till it turned into a rich fertilizer. Pesticides were used, especially the ashes of materials like cowdung, sesame, and even honey and ghee. Pastes of various kinds were employed to nurse sick plants back to health. These pastes were made of the leaves of neem, or seeds like the soapnut, mustard seed and linseed, or of gums like asafoetida. They understood that land needs to rest after growing one crop, and practised this. Crop rotation and seasonal sowing was very well understood indeed. Rice was sown in the rainy season and reaped at the onset of the following summer. Then came lentils, beans and peas, harvested in spring, while

barley, wheat and linseed were sown in winter and reaped before summer commenced. The wide variety of foodstuffs is shown in a prayer by these pastoral people, given in the *Yajurveda* and composed about 800 BC:

May for me prosper, through the yajna (sacrifice), milk, sap, ghee, honey, eating and drinking at the common table, ploughing, rains, conquest, victory, wealth, riches. May for me prosper through the yajna, low-grade food, freedom from hunger, rice, barley, sesame, kidney beans, vetches, wheat, lentils, millets, pascum grains and wild rice. May for me prosper, through the yajna, trees, plants, that which grows in ploughed land, and that which grows in unploughed land.

These crops were all grown in fields, and the outskirts of villages were used to raise supplementary food like fruits and vegetables. Banks of rivers were considered suitable for growing pumpkins and gourds, and lands which frequently overflowed with water for pepper, grapes and sugarcane. Vegetables and root crops were raised in the vicinity of wells. The moist beds of lakes were well-suited to green crops. Marginal furrows between two rows of other crops were recommended for planting fragrant plants and medicinal herbs.

Ways of cooking

Barley, the earliest Aryan grain, was fried and made into cakes and dipped in ghee for eating. Barley-flour balls coated with honey furnished a delicious sweetmeat. Rice was cooked with water or milk, or eaten along with curds, ghee, sesame (*til*) seed or meat. Puffed rice was known and eaten either as such, or as a drink after boiling with milk. What we call *chivda* was then called *prthuka*, and it was then made by flattening dampened rice grains with a pestle exactly as is done now. A dish of several cereals cooked together was called *chitranna*. The three common pulses were the three m's: *masha* (*urad*), *mudga* (*mung*), and *masura* (*masoor*). Soaked, ground and fermented dhal, especially *masha*, was made into fried materials called *vatakas*, which are the *vadas* of today. Sweet rice dishes like *kshirika* (*kheer*) were made using milk and jaggery. Pulses were eaten as they still are in the form of soups, *vatakas*, and *parpatas* (*papads*).

Milk products were extremely popular. Milk was curdled to give *dadhi* (curds), now called *dahi*. Curds mixed with fresh milk was a popular refreshing drink. Milk solids were precipitated with acidic lime-juice, then pressed to *dadhanwat* (*paneer*). Ghee was very widely used for frying purposes. The modern *shrikhand* was then called *shikharini*, and was sweetened with sugar and spice. Buttermilk was widely drunk. An ingenious method was used to carry milk: a cloth was soaked in milk, dried, and the adhering powder reconstituted whenever needed with hot water.

We read about many vegetables, like lotus stems and roots, several other water plants, the bottle gourd and cucumber, the bitter gourd or *karela*, many yams and

tubers, and a large number of green leafy vegetables like *palak* (*keerai*). Fruits included the date, mango, jamun, coconut, banana and jackfruit, besides figs, grapes, oranges, citrons and limes. Many forest fruits, used by hermits who lived there, are listed as well.

Though ghee was preferred by the Aryans for frying, the common population used sesame oil, mustard oil, coconut oil and several others.

There were also a large number of spices to be had. These included the turmeric, mustard seed, lime, long pepper, round black pepper, asafoetida, cumin seed (*jeem*), ginger, cloves and many kinds of vinegar. Fruit juices were spiced to serve as refreshing drinks. Sour vinegar was made from many sweet materials like honey, sugar and jackfruit. Another very simple soured material was rice-water that had been left overnight to turn acidic by fermentation. In fact, with a little pepper, salt and spices added, this was a popular morning drink.

Of course there were many kinds of meat. We read of mutton, deer meat and pork, cooked with fruit juices.

Honey was the first material used by human beings for sweetening. There are very old cave paintings which show honeycombs being disturbed by men, and the latter being badly bitten for their desecration! Since India grew sugarcane from an early date, it also developed products like jaggery, sugar candy, and later crystal sugar. Just like today, jaggery or *guda* (*gur*) was often used in making laddus from sesame seed, puffed rice, roasted wheat, fried barley and so on. Many sweets were made of envelopes of flour stuffed with fillings which included jaggery, shredded coconut or copra, sesame seeds and so on, and then fried in ghee. Puffs, *samosas* and *modakas* were developed then, and are extremely popular even today. Rice was cooked with milk, sugar and ghee to give *payasa*, the same word that is used even now.

Salt of several kinds was known. Sea salt was one, rock salt another, and black salt yet another.

Drinking water was always important in the Aryan way of life. It was also realized that water was frequently contaminated. Rainwater was particularly recommended for its purity. Special tanks were built in which lotus plants were grown to purify the water. The boiling of water and the boiling of milk, which we still carefully practice today in our homes, were a strict part of the food habits of the ancient Aryans.

Food variety

Even in those far-off days, nearly 3 000 years ago, quite an extraordinary variety of all kinds of food was available to the people. Many cooking practices which we still pursue have come to us from these ancient ancestors of ours. In fact, it is true to say that the only food materials which came to us later on did so after India became exposed to the influence of Europe. In the 16th and 17th centuries AD, Portuguese and Spanish seafaring people brought into India food plants from South and Central America. What these are, we will learn about later. To whet our appetites, let us just say at this point that chillis were one of the foods which came to us centuries later from Mexico.

OceanofPDF.com

5 THEORY AND PRACTICE

Sanitation and hygiene

Let us continue our story of the Aryan way of life, because its influence was strong enough to persist even to this day. The kitchen was considered the seat of ritual, and was in fact located next to the *puja* room. The cook was expected to have a bath, change into fresh clothes and only then enter the kitchen. Once cooking had started, he could not leave the kitchen till it was over; if he did so, he would be considered polluted. The inmates of the house ate in the kitchen and had to wash their hands, feet and mouth before sitting down on the floor to eat. Of course, after they had finished, there was a similar thorough washing of the mouth and the hands. Though this was considered as a part of the ritual, it was obviously aimed at cleanliness and good hygiene. They ate food off disposable plates made of leaves. A common one was the large banana leaf, but there were others made by joining together leaves of the banyan or the *palas* tree using small splinters of bamboo. Even cups were made by folding up these leaves and using bamboo-splinter hooks in ingenious ways. After the meal, these were all thrown away so that no pollution passed from one person to another. Water could not be sipped but had to be poured into the mouth, so that the lips did not come in contact with the rim of the cup used. There was a strong belief against eating food that was even slightly stale. For example, if two meals were eaten in a day, no material from the first meal could be eaten at the second; everything had to be cooked afresh, especially boiled food. This was obviously because of the fear of spoilage of food, which of course is very rapid in hot climates, and can cause serious infections. All water that was drunk was boiled and then cooled, and milk also was never used before it had been thoroughly brought to the boil. In these semi-religious instructions, one can see over and over again the emphasis our ancestors placed on good hygiene and good sanitation.

Cooking concepts

In the orthodox Aryan rituals, two kinds of food were distinguished. One was called *kachcha* food and the other *pucca* food. *Kachcha* food consisted of things like boiled rice, chappatis, and boiled dhal, which were common everyday foods cooked in the kitchen and eaten by the family at every meal without leaving the premises of the kitchen. *Pucca* foods were those fried in oil, and these foods could be shared with non-family members without incurring any pollution. Today, we realize that fried foods are much less liable to contamination with micro-organisms like bacteria, and can therefore be kept for quite a long period after being made. *Chivda*, *murukku* and *boondi* are even carried on long journeys, whereas rice or chappati would be inedible in a day or two. Frying was, therefore, an important early concept in the Aryan way of life. Great sanctity was attached to the use of ghee for frying. The Aryans looked down upon any other cooking oils, and insisted that everything should be fried in pure ghee. The older inhabitants of India had used oils like those of the sesame, mustard and coconut, and gradually these also began to find a place in Aryan cooking for special purposes to impart their own flavours. But the high value placed on ghee still persists in India, wherever one can afford to buy and use ghee. Of course it does have a very pleasant flavour, and also gives structure to certain sweets like *Mysore pak* which cannot be had with vegetable oils.

Ethical ideas

The Aryans placed a very high value on food. One of the *Upanishads* says:

Man consists of the essence of food. From food all creatures are produced, and by food do they grow.

Food was not simply for sustenance, or taste, or nutrition, but was believed to have certain spiritual qualities. The purpose of food was to join together the spiritual aspirations of the person eating the food with the higher purpose of living. Food was part of a large, moral, cosmic cycle in which the spiritual part of human nature had to be sustained as much as the bodily functions of living and breathing. While many of these theories may be considered fanciful, it did lead to an exalted place being given for food in the entire business of living and of one's ultimate purpose.

Hospitality was another part of these ethical ideas. It was believed that the highest merit went to someone who shared his food even when he had just enough for himself. Guests had to be treated with great honour. Among those who had to receive high hospitality were widows, any relatives, and even animals like dogs and crows. Giving with a good heart was the moral behind numerous stories and verses, and was considered a virtue in itself. The giver was indeed blessed.

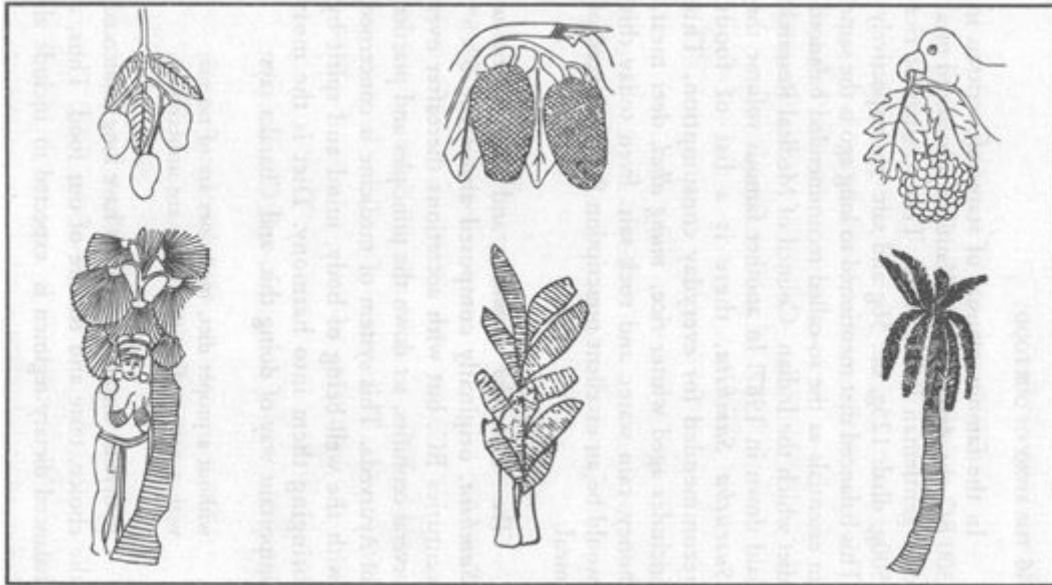
Nutritional and medical concepts

A number of practices were developed in regard to food preparation which are now realized to be of high nutritional value. Thus, from very early days, the parching and puffing of grains like rice, wheat and barley was practised. Today, we know that such processes break down carbohydrates and proteins and makes them easier to digest. In fact, puffed rice can be boiled with milk and is easily digestible even by a child. Another early concept was that of sprouting: grains like wheat, ragi, green gram or channa were soaked in water and wrapped in a wet cloth till they started to sprout. After two or three days, when the sprouts were about a centimetre long, they were used as food. Scientists have shown that several vitamins, increase enormously in content during the sprouting process: vitamin C goes up ten times, and several vitamins of the B-group are doubled. Also, the iron constituents present are broken down into simpler forms and become available to the body. Sprouted grains now have a very high place in nutritional diets in Western countries, so it is interesting to find that they were used in India as far back as 3 000 years ago. Pickling, again a very old Indian practice, conserves vitamins for use when fresh food is not available. Two excellent examples are pickled lime and pickled amla (nellikai), which are both rich in vitamin C.

Another good practice handed down from those Aryan times was the use of plenty of green leafy vegetables at every meal. Green leaves are extremely rich sources of vitamins and minerals, and nutritionists of today recommend their consumption for this purpose. Another fine concept developed long ago was the blending of differing types of proteins by using cereals and *dhals* together. It is well known now that this leads to a higher quality of the total proteins from both sources, higher than in either the cereal or the pulse taken separately. What are some examples of such blended foods? One is *khichdi*, in which rice and pulses are cooked together. The *dosa* of south India uses rice and *urad dhal*, and in addition the mixture is fermented so that it becomes far more easily digestible and also more nutritious. The *idli* is a similar fermented and steamed food, and in western India there are equivalent products called the *dhokla* and *khaman*.

The copious use of fruits and vegetables was another admirable feature of Indian food from early times. The Buddhists especially stressed this aspect, and even their stories and their sculptured stupas have delightful depictions of the abundant flora of India (see Fig 5).

Fig. 5: Flora Depicted in Buddhist Sculpture



In the famous manual of statecraft written in 300 BC, the *Arthashastra* of Kautilya, a balanced meal of a gentleman is described. This consists of rice 500g, dhal: 125g, oil: 56g and salt: 5g, respectively. This balanced diet mentioned so long ago is the same in essentials as the so-called recommended balanced diet which the Indian Council of Medical Research laid down in 1987! In another famous volume the *Susrutha Samhita*, there is a list of foods recommended for everyday consumption. This includes aged winter rice, *mung dhal*, deer meat, honey, rain water, and rock salt. Even today this would be an excellent prescription for a balanced of meal.

The *Charaka Samhita* and the *Susrutha Samhita*, originally composed about the 6th/5th centuries BC, but with accretions thereafter over several centuries, set down the principles and practice of Ayurveda. This system of medicine is concerned with the well-being of body, mind and spirit by bringing them into harmony. Diet is the most important way of doing this, and Charaka says:

without a proper diet, medicines are of no use;

with a proper diet, medicines are unnecessary.

Ayurvedic dietary injunctions have long influenced the choice, taste and texture of our food. Thus, a balanced dietary regimen is expected to include all the six major tastes, namely sweet, sour, salty, astringent, pungent and bitter. Textural variety should be had by including foods meant for licking, sucking, chewing and drinking. Even today this will be seen in a full traditional wedding meal of your community and region. Think about this.

Vegetarianism

The Aryans ate meat of all kinds as we have seen. A meal is described in the *Ramayana* as follows:

Cleanly cooks, under the supervision of diligent stewards, served large pieces of meat roasted on spits; meats cooked as curries and sauces made of tamarind and pomegranate; young buffalo calves roasted on spits with ghee dropping on them; the same fried in ghee, seasoned with acids, rock-salt and fragrant leaves; large haunches of venison boiled in different ways with spices and mangoes, and sprinkled over with condiments; shoulders and rounds of animals dressed in ghee, sprinkled over with sea-salt and powdered black pepper, and garnished with radishes, pomegranates, lemon, fragrant herbs, asafoetida and ginger.

However, even from the very early Aryan period, there was a feeling of unease expressed at the use of beef as food. The argument was mainly an economic one. Cows were useful on the farm, and they gave so many good things to human beings, like milk, urine, dung and leather. It was realised that valuable resources were being sacrificed when one killed a cow just to eat it. In the beginning the prohibition was mainly against cows rather than bulls, but later on cattle meat as a whole began to be given up. Gautama Buddha, for example, was very opposed to killing for sacrificial purposes or for eating, and his humanitarian views brought many Hindus into the Buddhist fold. Another great religious leader, Mahavira, was even stronger in this regard, and his followers, the Jains, became very strict vegetarians. These trends led Hinduism also to respect vegetarian sentiments. In place of the animal head used in the past in ritual sacrifices, vegetable materials such as coconuts and pumpkins came to be used. Gradually, the Brahmins became total vegetarians and gave up all forms of animal food. It is interesting to remember that it was only in India at that time (about 600 BC) that a totally vegetarian diet was even possible, because of the very large number of cereals, pulses, vegetables and fruits that were available in the country. In most other countries then, it would have been almost impossible to be a vegetarian. Even today, when we leave our own country and go abroad, we realise that to be a vegetarian outside India leads to having very dull meals.

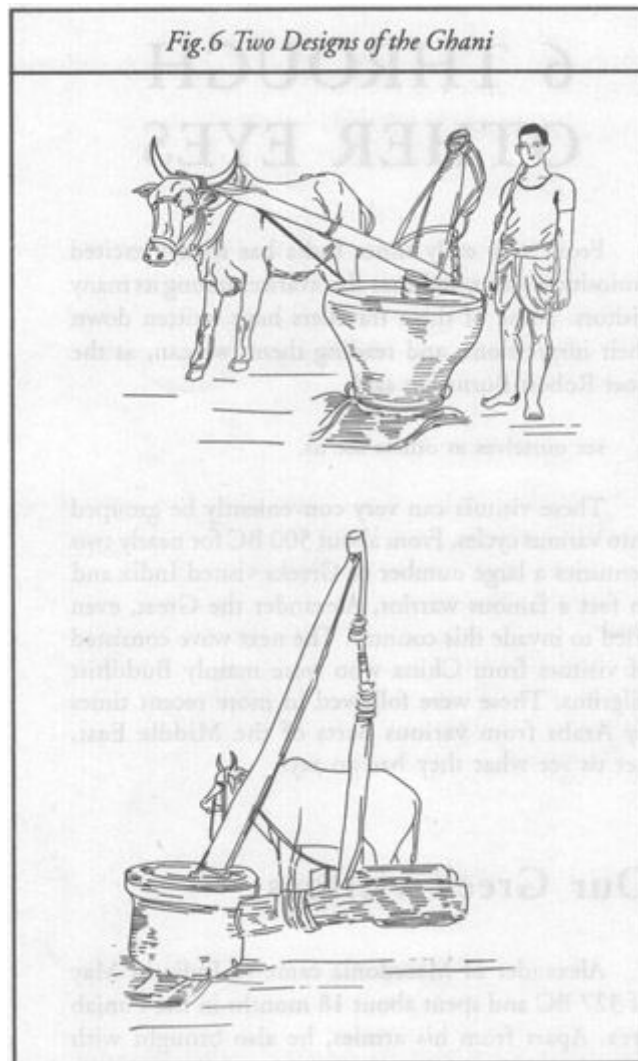
Appliances

Somehow the Indian mind has always been quite content with simple appliances to do manual jobs. Even in the Indus Valley civilization, we saw that grinding-stones and saddle-querns were used, exactly as they still are in our kitchens. They have remained unchanged for a very long span of time. Pounders for grain using the hands or feet are just the same as they were thousands of years ago. So is the *shurpa* or *morram*, made of bamboo strips, for removing sticks and stones from grain. Deep-frying vessels found in Mohenjodaro and Harappa are exactly like those which we use even today. Tandoori ovens have been found in these cities which are identical with those of the present. In certain other cities, a cooking *chula* with three raised knobs to place the vessel on, has been found, exactly like those now in use. Portable metal *sigris* have

been around for thousands of years. Cooking vessels of various functional shapes, like frying pans, have more or less remain unchanged over centuries.

One of the more sophisticated appliances that has come down to us is the one used for crushing oil from oilseeds. It is called the *ghani* or *kolhu* in the north, and the *chekku* in south India. A yoked animal goes round and round, crushing the oilseed in a kind of mortar-and-pestle arrangement. The *ghani* seems to have sprung from devices for pounding grains and for crushing sugarcane which must have been known from very early times. It is quite sophisticated in its construction and principles and is, therefore, unusual among the appliances for food-processing and cooking that have been passed down to us from ancient times (see Fig. 6).

Since the turn of the century, the number of *ghanis* in use in India has come down rapidly, to be replaced by larger modern technologies.



6 THROUGH OTHER EYES

From very early times India has always excited curiosity, wonder, and later also avarice, among its many visitors. Some of these travellers have written down their impressions, and reading them, we can, as the poet Robert Burns has said

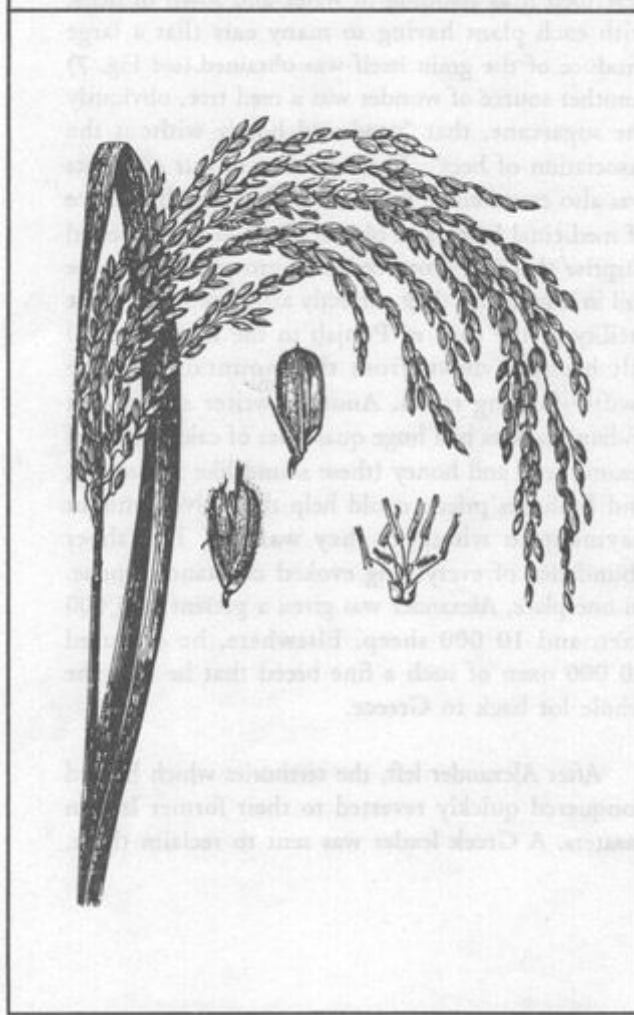
see ourselves as others see us.

These visitors can very conveniently be grouped into various cycles. From about 500 BC for nearly two centuries a large number of Greeks visited India and in fact a famous warrior, Alexander the Great, even tried to invade this country. The next wave consisted of visitors from China who were mainly Buddhist pilgrims. These were followed in more recent times by Arabs from various parts of the Middle East. Let us see what they had to say.

Our Greek visitors

Alexander of Macedonia came to India in May of 327 BC and spent about 18 months in the Punjab area. Apart from his armies, he also brought with him a team of writers and "philosophers", as scientists were then known. Several of them wrote accounts of their visit. Rice was to them a strange plant; they described it as standing in water and sown in beds, with each plant having so many ears that a large produce of the grain itself was obtained, (see Fig. 7) Another source of wonder was a reed tree, obviously the sugarcane, that "produced honey without the association of bees". The large size of our coconuts was also commented upon, as well as the abundance of medicinal herbs and plants. The Greeks expressed surprise that two crops could be grown on the same soil in a year, and they correctly attributed this "great facility of the soil" of Punjab to the fertile alluvial silt brought down from the mountains by the swiftly-flowing rivers. Another writer stated that Indian markets had huge quantities of cakes made of sesame seed and honey (these sound like *til laddus*), and Brahmin priests could help themselves without payment to whatever they wanted! The sheer abundance of everything evoked constant surprise. In one place, Alexander was given a present of 3 600 oxen and 10 000 sheep. Elsewhere, he captured 20 000 oxen of such a fine breed that he sent the whole lot back to Greece.

Fig.7 The Rice Plant Astonished the Greeks



After Alexander left, the territories which he had conquered quickly reverted to their former Indian masters. A Greek leader was sent to reclaim them, but was defeated. He left behind an ambassador called Megasthenes at the court of the Emperor Chandragupta Maurya. In fact, the Emperor even married a daughter of Megasthenes! Writing in about the year 320 BC, the ambassador was deeply impressed with the magnificence of the court of the Emperor, and the splendour of his palace. The Indian people he described as being frugal, simple in their manners and living in contentment. Rice was by far the common food, and rice-water was also drunk as a beverage. Agriculturists he talked about as

a most mild and gentle people ... who pursue their labours free from all alarm.

Rich Indians ate at a small three-legged table from a golden bowl (possibly a *thali* is referred to) in which was first placed rice, and then many other dainty accompaniments.

The Chinese pilgrims

Buddhism had been adopted quite widely in China, and there were many Buddhist monks anxious to see the places associated with the birth, the preaching and the death of the Buddha. They also wished to study the Buddhist sacred books abundantly available in India and to carry, if possible, a relict or two of the great teacher back to China with them. Many of these devout Chinese Buddhists lived in India for several years and wrote books which throw a very clear light on Indian life at that distant period of time.

One of the first of these was here for 15 years between 399 and 414 AD; this was Fa-Hsien, who was fortunately for us also a prolific writer.

No one kills any living thing, nor drinks wine, nor eats onion or garlic ... In this country they do not keep pigs or fowls, there are no dealings in cattle, no butchers' shops or distilleries in their market-places.

Of course this does not appear likely to be totally accurate, but may only mean that meat and drink was sold outside the city gates, as we know from other descriptions. He goes on:

The elders and gentries of these countries have instituted in their capitals free hospitals and hither come all poor or helpless patients, orphans, widows and cripples. They are well taken care of, a doctor attends them, food and medicine being supplied according to their needs.

Another visitor, I-Ching, spent as much as 24 years in India between 671 and 695 AD. He described the food of the Buddhist monks in the following terms. Guests were offered one of the eight syrups prescribed by the Buddha. At a meal, monks were first served two pieces of ginger with some salt, next came boiled rice, on which was poured a thin extract of beans and hot ghee; these were mixed with the fingers, after which cakes, fruits, ghee and sugar were served. Toothpicks were provided after the meal, followed by pure water for rinsing their mouths and sometimes a perfumed paste with which to cleanse their hands. The beverages that accompanied the meal were cold or warm water, whey, buttermilk or fermented, sour rice-gruel. Betel leaves carrying fragrant spices were served at the end to help digestion, remove phlegm, and leave the mouth fragrant.

Perhaps the greatest of all these learned Chinese monks was Xuan Zang, who spent 16 years between 629 and 645 AD in this country. He arrived after a most strenuous journey, starting from China, going across the barren and mountainous region of Tibet till he reached the monasteries in Afghanistan and Turkestan where he halted. He then came through the mountain passes of north-west India into Punjab. He describes in his famous travel book, called *Si-yu-ki*, these arduous journeys. In India itself he travelled so widely that he could write about visiting "110 of the 138

kingdoms" in every part of the country. Xuan Zang has this to say of the common people:

The Kshatriyas and the Brahmans are clean-handed and unostentatious, pure and simple in life and very frugal. They are pure of themselves and not from compulsion. With respect to the ordinary people, although they are naturally lightminded, yet they are upright and honourable. In money matters they are without craft, and in administering justice they are considerate. They are not deceitful or treacherous in their conduct, and are faithful to their oaths and promises. In their rules of Government there is remarkable rectitude, while in their behaviour there is much gentleness and sweetness.

And he goes on:

They are very particular in their personal cleanliness, and allow no remissness in this particular. All wash themselves before eating; they never use that which has been left over; they do not pass the dishes. Wooden and stone (clay) vessels, when used must be destroyed, vessels of gold, silver, copper, or iron, after each meal must be rubbed and polished. After eating they cleanse their teeth with a willow stick, and wash their hands and mouth. Until these ablutions are finished they do not touch one another. Every time they perform the functions of nature they wash their bodies and use perfumes of sandalwood or turmeric.

What fruits did Xuan Zang find in India?

The fruit of the amla, the *madkuka* (mahua), *bhadra* (ber), *kapittha* (wood-apple), *mocha* (plantain), *narikela* (coconut), and *panasa* (jack fruit) are common (Sanskrit words are used for these). It would be difficult to enumerate all the kinds of fruit; we have briefly named those most esteemed by the people. The pear, wild plum, peach, apricot, grape, etc. (Chinese terms are employed here) have all been brought from the country of Kashmir, and are found growing on every side. Pomegranates and sweet oranges are grown everywhere. The date, the loquat, the chestnut and persimmon are not known.

Elsewhere in the book the other products food are described.

In cultivating the land, those whose duty it is sow and reap, plough and weed, and plant according to the season; and after their labour they rest a while. Among the products of the ground, rice and corn (barley?) are most plentiful. With respect to edible herbs and plants, we may name ginger and mustard, melons and pumpkins, and others. Onions and garlic are little known, and few people eat them; if anyone uses them for food, they are expelled beyond the walls of the town. The most usual food is milk, butter, cream, soft sugar, sugar candy; the oil of the mustard seed, and all sorts of cakes made of corn are used as food. Fish, mutton, gazelle, and deer they eat mostly fresh, sometimes salted; they are forbidden to eat the flesh of the ox, ass, elephant, horse, pig, dog, fox, wolf, lion, monkey and all the hairy kind. Those who eat them are despised and scorned, and are universally reprobated; they live outside the walls, and are seldom seen among men.

With respect to the different kinds of wines and liquors, there are various sorts. The juice of the grape and sugarcane are used by the Kshatriyas as drink; the Vaisyas use strong fermented drinks; the Sramans and Brahmans drink a sort of syrup made from the grape or sugarcane, but not of the nature of fermented wine. The mixed classes and base-born differ in no way (as to food and drink) from the rest, except in respect of the vessels they use, which are very different both as to value and materials. There is no lack of suitable things for household use. Although they have saucepans and stewpans, yet they do not know the steamer used for cooking rice. They have many vessels made of dried clay; they seldom use red copper vessels; they eat from one vessel, mixing all sorts of condiments together, which they take up with their fingers. They have no spoons or cups, and in short no sort of chopsticks. When sick, however, they use copper drinking cups.

Xuan Zang spent about five years at the University of Nalanda in Bihar, not far from Patna. Students there were generously supplied with the four requisites of

clothes, food, bedding and medicine. Though their family be in affluent circumstances, such men make up their mind (after their studies) to be like vagrants and get their food by begging as they go about. With them there is

honour in knowing the truth and there is no disgrace in being destitute.

Xuan Zang's disciple, Shaman Hwui Li, has recorded his master's daily diet during his stay at Nalanda:

Every day he received 120 *jambiras* (citrus fruits), 20 arecanuts, 20 nutmegs, a tael (about 30 grams) of camphor, and a shang measure (about six kg perhaps) of *mahasali* rice. This rice is as large as the black bean and when cooked is aromatic and shining, like no other rice at all. It grows only in Magadha, and nowhere else. It is offered only to the king, or to religious persons of great distinction, and hence its name *mahasali*, or in Chinese *kung-jatin-mai* (rice offered to the great householder). He was also supplied every month with three tun (perhaps 3 kg) of oil, and as regards milk and butter he took as much every day as he needed.

The Arab influx

From about 850 AD, a steady stream of Arab visitors came into India from various parts of the Middle East. At that time, the province of Sind was under Muslim rule and this was the usual starting point for the journey. One of these visitors was Al-Masudi, a native of Baghdad who was here in 915-916 AD. He noted several prohibitions among the ordinary people against eating cows, tame poultry, and all kinds of eggs.

He also noted the huge amount of trade between India and China, to which ships sailed laden with Indian merchandise.

Al-Biruni, who was here for 13 years between 1017-1030 AD, was a learned man who tried hard to study and understand the ways of Indian life. In his famous book of travels called the *Rehala* he talks about the prohibition against eating beef:

As for the economical reason, we must keep in mind that the cow is the animal which serves man in travelling by carrying his load, in agriculture in the works of ploughing and sowing, and in the household by milk and the products made from it. Farther, man makes use of its dung and in winter-time even of its breath. Therefore, it was forbidden to eat cow's meat.

Of the eating habits of the Indians he wrote:

The Hindus eat singly, one by one on a tablecloth of dung. They do not make use of the remainder (remnants) of a meal, and the plates from which they have eaten are thrown away if they are earthen. They have red teeth on account of chewing areca nuts with betel leaves and lime. They drink wine before having eaten anything, then they take their meal. They sip the stall (urine) of cows, but do not eat their meat.

Elsewhere Al-Biruni states that wine

is allowed to the Sudra. He may drink it but dare not sell it (just) as he is not allowed to sell meat.

The rules prohibiting the killing of animals

applies in particular only to Brahmins, because they are the guardians of the religion, and because it forbids them to give way to their lusts ... It is allowed to kill animals only by strangulation, but only certain animals, others being excluded... Animals of which the killing is allowed are sheep, goats, gazelles, hares, the rhinoceros, the buffalo, fish, water and land-birds such as sparrows, ring-doves, francolins, doves, peacocks, and other animals which are not loathsome to man or noxious.

The Moroccan, Ibn Battuta, was in India for several years around 1340 AD. He was astonished to watch the removal, from the outer walls of Delhi Fort, of grain which had been stored there for 90 years, but was still in good condition. He also noted, with approval, the rotation of crops:

After the spring crops of wheat, barley, *channa* and lentils had been harvested, in the same land the autumn crops were grown for their land is generous and of good heart.

As for rice, he noted with astonishment that it was sown and harvested even three times a year, and was the principal crop of the country. Sesame seeds and sugarcane were also grown as autumn crops. He made particular mention of fruits which were strange to him. One of these was the mango, and another was the jackfruit which he described as "the best fruit in India". He noted jamuns and pomegranates, and stated that grapes were

very rare in India and are to be had only in certain districts, and in the capital, Delhi.

Elsewhere in his book he mentions that grapes were grown in Daulatabad. The best sugarcanes, in his opinion, came from Kerala. The coconut intrigued him: it

resembles a man's head for it has marks like eyes and a mouth, and the contents when it is green are like the brain. It has fibre like hair.

With the vast productivity of Bengal, he was much impressed:

Nowhere in the world,
he wrote,

have I seen any land where prices are lower than there.

OceanofPDF.com

7 THE MUSLIM CONTRIBUTION

From about 700 AD, and continuing for nearly 300 years, small Arab Muslim kingdoms existed in the province of Sind, which is now part of Pakistan. In the next millenium, for nearly two centuries, warlike chieftans like Mahmud Ghazni and later Muhammad Ghorī made repeated raids into India with the sole intention to plunder and rapture rich booty. And then from 1206 AD came the actual occupation of north India, and the setting up in Delhi of what are now called the Sultanates. The first of these was the Slave Dynasty, then came the Khaljis and the Tughlaks, and these were followed by the Sayyids and the Lodis. In 1526, on his fifth expedition into India, Babar won the Battle of Panipat, and established in Delhi the Mughal dynasty. For the next two centuries, under six emperors, a magnificent empire was established whose glory became known far and wide.

This long interaction of people of the Muslim faith with India influenced many things in this country, including our food. Of course, the ingredients that were used remained more or less the same. What really changed were the ways of using them and the styles of cooking. Also, the Muslim rulers set the style for an elaborate code of table manners which was different from the prevailing Hindu ethos. Another significant contribution of the Mughal rulers to India was the systematic cultivation of fruit orchards. With their passion for good fruit, peaches, pears, cherries, grapes, melons and mangoes were cultivated with great diligence and became extremely abundant. Eating the first melons and grapes of fine varieties that he had caused to be raised in Hindusthan, Babar wrote,

filled my measure of content.

Food items

Even items of food that were known already in India became enriched and almost transformed. From very early times, *palao* was known in India, and was referred to in Sanskrit by this very name, or as *mamsaudana* (meat-rice). In fact, we read in the Ramayana that the favourite food of Sita was a *palao* made of deer meat. However,

the *biriyani*, *shulla* and *shurba* of the Muslim cuisine were more elaborate. The meat was cooked till it was almost about to disintegrate and become part of the rice. Spices like turmeric, pepper and cumin seed (*jeera*) were included, and there was a free use of strong aromatic spices called *garam masala*, which included cloves, cardamom and cinnamon. Exotic ingredients like the stamens of *zafran* or saffron (*kesar*), with their beautiful colour and strong fragrance, were freely employed. Various nuts like almonds and pistachios were used, and raisins were scattered throughout the rice. The word *biriyani* itself is derived from *birinj*, which is simply the Persian word for rice.

Roasting pieces of meat on spits over an open fire have long been described in Sanskrit literature, for example in the *Ramayana*. The Muslim culinary genius transformed this into numerous varieties of *kababs* like *seekh kabab* and *shammi kabab*, each with its own texture and flavour. A cutler called *shikampoor* was shaped from meat that had been ground into a really fine paste, with the spices also fully blended into it: this small cutlet literally melts in the mouth. *Samosa* was likewise a traditional Indian dish which was similarly exalted, the meat stuffing being enriched with walnuts, pistachios, onions and spices. The Muslims avidly adopted the use of ghee as a cooking fat and used it freely in making all these dishes.

A sweet tooth was not neglected. *Halwas* were made of wheat flour, or from the pulps of various fruits enriched with sugar, nuts, ghee and silver foil. We read in the early Muslim writings of something called *khisti*, a confection of white flour, sugar and ghee, and of *subunia* made of sesame seeds, almonds and honey. Sweet drinks, usually heavily flavoured, were also common. There were many favoured *sherbets*; one called *fuqqa* was based on barley water.

The ritual of dining

Large public dinners were a part of court etiquette, and such repasts were even held twice a day. We have a vivid account of them by the Moorish traveller, Ibn Battuta; he lived in India for nearly 20 years from 1325 AD, a period which coincided almost exactly with the rule of Sultan Muhammad bin Tughlak. This is what he writes:

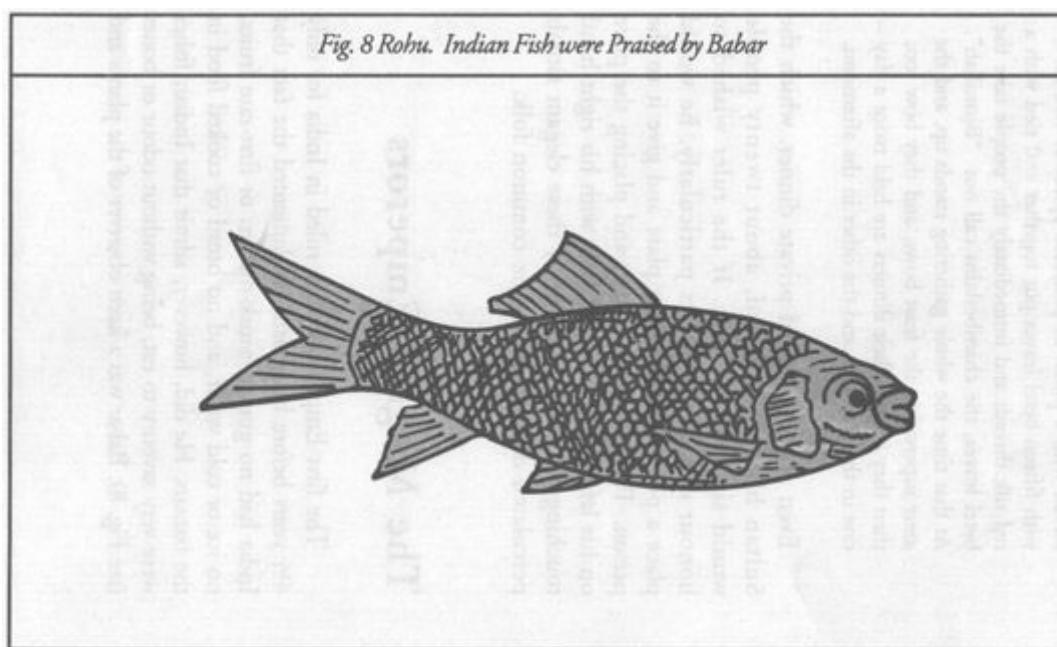
There custom is that the judges (*qazis*), orators (*khatibs*), jurists (*shorfa*), *saiyids* and dervishes (*mashaikh*) sit at the head of the dinner carpet (*simat*): and then come the Sultan's relatives, the great *amirs* and the rest of the people. But none sits except at Iris appointed place: and thus there is absolutely no confusion amongst them. All having taken their respective seats, the cup-bearers (*shurbdariya*) who give the drink come holding in their hands gold and silver and copper and glass vessels filled with refined sugar dissolved in water, which they drink before the dinner. As they drink the chamberlains (*hujjab*) call out "Bismillah", then they start eating. Every one has before him a set of all the various dishes comprising the dinner, which he eats exclusively; and no one shares his plate with another. When they finish eating, the drink (*fuqqa*) is served in pewter tankards; and as soon as the people take it the chamberlains call out "Bismillah". Then are brought trays full of betel leaves and spices, and everyone is served with a pinch of pounded spices as well as with fifteen betel leaves put together and tied

with a red silk thread; and immediately the people take the betel leaves, the chamberlains call out "Bismillah". At that time the whole gathering stands up, and the *amir* supervising the feast bows, and they bow too; then they retire. Their dinners are held twice a day -one in the forenoon and the other in the afternoon.

Even at the so-called private dinner, which the Sultan himself attended, about twenty people would sit down to dine. If the ruler wished to honour one of the guests particularly, he would place a piece of bread on a plate and give it to the person. The latter received it: and placing the plate on his left hand, he would bow with his right hand touching the ground. Some of these elegant rituals percolated down even to the common folk.

The Mughal Emperors

The first Emperor, Babar, ruled in India for only 41/2 years before his death. He lamented the fact that India had no grapes, musk-melons or first-rate fruits, no ice or cold water, and no bread or cooked food in the bazaars. He did, however, admit that Indian fishes were very savoury to eat, being without odour or bones (see Fig. 8). Babar was a keen observer of the plants and animals of his new country. For example, it is from his writings that we know that there were eight varieties of citrus fruit in India at that time, and these included the sweet orange, lime, citron and *santhra*.



His son, Humayun, was more Indianised, even arguing that beef was not a food for the devout. He even became a vegetarian for a while after taking a vow to win back his kingdom from the interloper Sher Shah.

The great Emperor, Akbar, did not like meat and only took it seasonally. In fact, he was practically a vegetarian, though his table was very sumptuous. Father Monserrate,

a Catholic priest, who spent some time at Akbar's court, reported that more than forty courses were served in great dishes at the royal table. These were wrapped in linen cloths, and tied up and sealed by the cook, for fear of poisoning before being brought to the table. Akbar's reign was beautifully chronicled by one of his courtiers, Abul Fazl. His famous book, called the *Ain-i-Akbari*, was written in 1590 and is a mine of information on many things of the time. Three types of dishes were described as being cooked in the royal kitchen. The first of these was *sufiyana*, which entirely consisted of vegetarian dishes like rice, wheat, dhals, green leafy vegetables, *halwas* and *sherbets*. The next group consisted of preparations of rice with meat, like *palaos*, *biriyani*, *shulla* and *shurba*, or of wheat with meat, like *haleem*, *harisa*, *kashkh* and *qutab*. Then came a group of purely meat dishes cooked with ghee, spices, curd and eggs, to give such rich items as *yakhni* (a Kashmiri speciality), *kabab*, *dopiyaza* (with plenty of onions), *dumpukt* and *qualiya*.

The *Ain-i-Akbari* also describes the passion of the Emperor for growing fruits, and the fine varieties of mangoes which the Portuguese had developed by grafting techniques, which were also avidly pursued by the Mughals. The food items available in the Delhi market are listed with their prices, and no less than 35 fruits are noted. He remarks:

Melons and grapes have become very plentiful and excellent; and watermelons, peaches, almonds, pistachios, pomegranates, etc. are everywhere to be found.

The list of fruits includes two surprising items, namely the pineapple and custard apple. These are products, native to South and Central America, which were unknown in India before the coming of Vasco da Gama in 1498. Yet less than a hundred years later they are on sale in the Delhi market, a remarkable example of quick transfer and adoption of food plants.

Akbar's son, Jahangir, had great curiosity about anything novel. He did not hesitate to experiment with eating any new edible vegetable or animal that was found in the country. He records in his memoirs that the flesh of the mountain goat was more delicious than that of all other wild animals, but that of the wild ass is

in no way suited to my taste.

He even went so far as to drink the milk of the female antelope; he found this palatable, and added:

They say it is of great use in asthma.

His favourite food was a *khichdi* of Gujarat called *lazizan* which was enriched with ghee, pistachios, raisins and spices. Another favourite of his was *faluda*, a rich jelly-like drink made from the strainings of boiled wheat mixed with fruit juices and cream.

The common people

Of courses, *faluda* and *lazizan* were not for ordinary folk. The poorer Muslim for his breakfast was fond of *naan*, the puffed roti made from *maida* after slight fermenting of the dough, followed by baking in an oven. It was generally eaten with *kheema* or *kababs*. For the main meal, rice was the favourite, eaten with plenty of onions. In fact, the free use of onions in Indian cooking seems to have been a Muslim contribution, because they were used either not at all, or very sparingly, by most Hindu cooks. The popular dessert of ordinary people was either threads of wheat (*pheni*) or rice cooked in milk and sugar (*kheer*). *Halwas* and dried fruits were also popular. Betel-chewing was adopted so avidly that it became, if anything, even more popular among the Muslims than the Hindus.

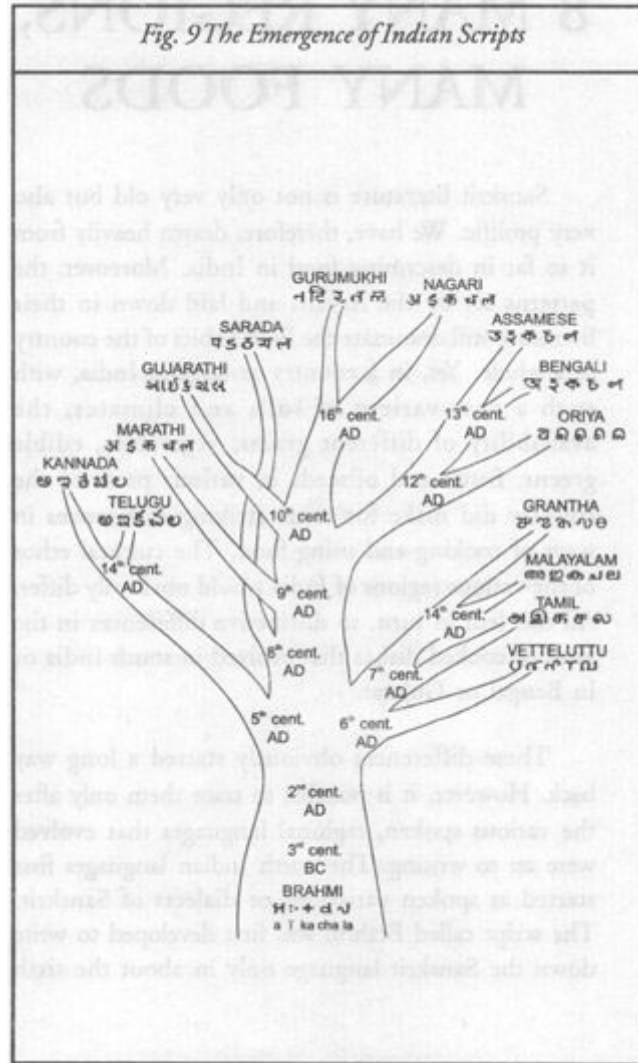
In using wheat, the preference of the Hindu was for fried items like *puris* and *bhaturas*, which were eaten with various vegetable accompaniments. Rice cooked with dhal (*khichdi*) was the common everyday Hindu meal, and very poor people simply ate boiled rice with a piece of green ginger for taste, or a little salt and pepper. Many farmers would carry with them to the fields parched and puffed grains like rice and gram, and munch them from time to time as they were working. Certain items were popular only in specific areas. We read of *rotis* made of *jowar* (sorghum) in Gujarat and of *bajra* (pearl millet) elsewhere. In south India, certain areas favoured balls of *ragi* flour which were eaten with a spicy accompaniment. We shall look into such regional food variations in the next chapter.

8 MANY REGIONS, MANY FOODS

Sanskrit literature is not only very old but also very prolific. We have, therefore, drawn heavily from it so far in describing food in India. Moreover, the patterns set by the Aryans and laid down in their literature still dominate the food habits of the country as a whole. "Yet, in a country as large as India, with such a vast variety of soils and climates, the availability of different grains, vegetables, edible greens, fruits and oilseeds in various parts of the country did make for some striking differences in ways of cooking and using food. The cultural ethos of the various regions of India would obviously differ. All this led, in turn, to distinctive differences in the kind of cooked dishes that evolved in south India or in Bengal or Gujarat.

These differences obviously started a long way back. However, it is possible to trace them only after the various spoken, regional languages that evolved were set to writing. The north Indian languages first started as spoken variations or dialects of Sanskrit. The script called Brahmi was first developed to write down the Sanskrit language only in about the sixth century BC. In course of time it was gradually modified for such languages as Bengali, Gujarathi and Marathi, which took a definite shape in writing about the 10th-11th centuries AD. Even the south Indian languages owe their scripts to Brahmi, starting with the Tamil Brahmi script which took shape in about the 3rd century BC to set Tamil poetry to writing. Kannada and Telugu emerged around the 5th century AD, and Malayalam some eight centuries later. It is writings in these regional languages that we shall draw upon as authentic sources to throw light on the distinctive food habits of the people who spoke (see Fig.9) and wrote in them.

Fig. 9 The Emergence of Indian Scripts



The Tamil country

In Chapter 3 we learnt how *ragi* and rice were the staple foods of south India from very early times, how *tuvar dhal* originated, and how the coconut and its oil were known a long time ago. We also learned that other foodgrains like *varagu*, *bajra* and *jowar* were used, while archaeological excavations in the south had shown that horsegram, *mung dhal* and *masoor dhal* had been known and eaten for a very long time indeed. In south India, even 4 000 years ago, huge herds of cattle were tended in pens. This also implies that dairy products were known and, indeed, the oldest Tamil writings have numerous words for cream, curds, butter, and ghee.

Many animal foods are described with great relish in the early Tamil literature.

Even Brahmins did not lack relish for the meat and toddy served to them at feasts held by the chieftans and princes of the land.

Beef was freely eaten; there are four names for this meat in the early Tamil language, showing that it was a common and well-liked food. In the north, as we have seen, the domestic fowl was not eaten, but there was no such taboo in the south. Other delicacies were the cooked *aral* fish served piping hot, and the meat of the tortoise, rabbit and hare. Wild boar was hunted using nets; it was then kept in a pit and fattened by feeding it with rice flour to yield pork of exceptional taste. Here is a description from the Tamil literature of a feast given about 150 AD by a Chola ruler:

Goblets of gold with intoxicating liquor, soft-boiled legs of sheep fed on sweet grass, and hot meat, in large chops, cooked on the points of spits ... fine cooked rice which, erect like fingers and with unbroken edges, resemble the buds of the *mullai* (jasmine) flower, together with curries sweetened with milk

It is interesting to note the reference to wine and to roast *kababs*, and the beautiful comparison of shining white rice grains to jasmine buds. Tamil literature also describes the brisk trade with both east and west from the ports of south India; one commodity brought in was Italian wine for use by the royalty.

The meat dishes cooked with pepper were called *kari* in Tamil, a word now used in English as curry. Fried spiced meat was called *tallita-kari*, fried meat was *pori-kari*, and meat with a sour sauce made of tamarind was termed *pulingari*. Even within the southern Tamil country there were quite marked differences in the food of different areas. This is beautifully shown in a passage from the *Perumpunnaru* written in the 1st century AD and describing the food that a wandering minstrel received at various places:

The hunters served him, on the broad leaf of a teak tree, with coarse rice of a red colour (hand-pounded perhaps, or naturally red?) and the flesh of the iguana. The shepherds gave him *cholam* (jowar) and beans and millet boiled in milk. In the agricultural tracts, the labourers invited him to a meal of white rice and the roasted flesh of the fowl. On the sea coast, the fisherman fed him with rice and fried fish in dishes made of palmyra leaves. The brahmin gave him fine rice with mango pickle, and the tender fruit of the pomegranate cooked with butter and the fragrant leaves of the *karuvembu* (*karipattha*). The farmers feasted him with sweetmeats, the fruit of the jack and the plantain, and the cooling waters of the coconut.

The main staple food was, of course, rice. Many forms were already known, like parched or flattened rice, puffed rice and parboiled rice. Rice *appam*, a pancake soaked in sweetened milk or coconut milk, was already well established, and *idi-appam*, cooked rice in the form of long white strings rather like fine noodles, is also mentioned even at this early date. Foods made from *dhal*, like the *dosai* and *adai*, were already in use. One food that does not appear is the *idli*, which is in fact first described not in Tamil, but in a Kannada work of the 10th century and a Sanskrit work of the 12th century. The *idli* today is a fermented mixture of rice and black gram *dhal* which is then steamed. The Chinese traveller, Xuan Zang, clearly states that in the 7th century Indians did not know the use of the steamer. There is a suggestion that the *idli* was developed in Indonesia, a part of which was then ruled by Hindu

kings. It is called *kedli* there, and the Indian cooks of these kings may have returned home with the recipe.

Bengal

Distinctive culinary habits have marked the food of Bengal from very early times. Even the priesdy class writing in Sanskrit, always argued for the right of the Brahman to eat fish. In contrast to the prohibitions against eating any form of flesh elsewhere, the Brahmans of Bengal continue, even to this day, to eat fish. Ramakrishna Paramahansa and Swami Vivekananda enjoyed fish dishes. From very early times, two fish species that are still popular are mentioned. One is *illisd*, the *hilsa* of today, and the other is *rohituka*, the *rohu* or *rui* of today (both, fresh-water fish).

A wide variety of vegetation has always characterised the well-watered and rich soil of Bengal. All visitors repeatedly remarked on the abundance of vegetables, edible green leaves, tubers of innumerable kinds, fruit and sugarcane. The Bengali bride takes as her example two deities—one is Draupadi, who was able to feed five husbands to their satisfaction in the *Mahabharata*, and the other is the supreme cook Ganga, the wife of King Shantanu and the mother of the noble Bhishma, who tutored both the Pandavas and the Kauravas. The distinctive flavours in a total meal which is enjoined in the Vedic texts is still put into practice with great finesse in Bengali cooking. Dishes with a bitter flavour are achieved through use of materials like the bittergourd (*karela*) or neem leaves, and this class of bitter dish has a special name 'shukto'. An astringent flavour is obtained by adding raw bananas, jackfruit seeds, or certain stems and stalks. The use of ginger juice, or of mustard seed or mustard oil, imparts to the dish a characteristic pungency which is an important flavour in Bengali cooking. Saltiness, another desired flavour, can of course arise from use of sea salt or rock salt, but also from certain naturally-salty green leaves. To obtain a sharp tart flavour, there are several choices such as green mangoes, tamarind, *amla* (gooseberry) fruit, lime, lemon or citron juice, or fermented rice water. Many fine distinctions are made between boiled, fully-fried, partly-fried, and roasted and fried dishes. There is also much play between dry and liquid items. Frying in ghee and frying in mustard oil are subtly employed. The Bengali has always had a sweet tooth. Luckily, some of the finest sugarcanes have always been raised in Bengal from historic times, and there was also palm jaggery with its distinctive and delicate flavour. Even early Bengali literature describes sweet dishes like *khiri* (*kheer*), *payesh*, *rabadi* (sweet, thickened milk), *manda* (a large *sandesh*) and *nadu* (*rabadi* with coconut shreds). Sweets made of rice flour were called *pitha*. Since those early times a host of sweets have arisen in Bengal based on *channa*, the solids precipitated from milk using an acid. Some of

these are *sandesh*, *gulab jamun*, and *lal mohan*, while the famous *rasogolla* and *ras-malai* were both invented in Bengal after 1850.

A poem, the ***Chandi Mangala***, written in the 16th century AD by Mukundrama Chakravarty, beautifully illustrates the traditional relationships between a person's emotional nature, the particular situation or status in life, and the food that is desired. In this poem Shiva, who is a strong, rather violent, *tamasic* character, is served with food that has been fried not in ghee but in mustard oil, and dishes that are bitter, pungent and acidic. On the other hand, Vishnu, a *gentle*, *sattvik* character, is served with ghee-fried foods and a number of delicate milk-based sweets. The pregnant wife of the hunter in the poem has no ghee and only roasted food, while the merchant's pregnant wife is given food with a tart flavour, rice but no tubers, and a variety of green herbs. Each food-spread or menu reflects both the person's character and situation in life.

Gujarat

The Gujarat area was much influenced by Jainism, which originated from about the 5th century BC, and enjoined very strict vegetarianism. Saints like Vallabhacharya also turned the people to vegetarian habits. Today, Gujarat has the highest proportion (nearly 70%) of people who are vegetarian. Even early Gujarati writings hardly mention meat dishes at all. Cooked rice was called *kur*, and cooked rice mixed with curds was called *karambo*. *Khandvi*, the thin, savoury pancake made of pulses, which is still popular in Gujarat, is described. Ghee, then as now, was plentifully used in cooking. In fact, *dhal* was cooked in ghee and so were *papads* and *badi* (*wadian*). Cooking oil is hardly ever mentioned. *Vedhami* was a sweet preparation based on *dhals* and wheat. Broken wheat with jaggery was called *sukhadi*, still an extremely popular dish. There was a light pastry described as *khaja*, besides *laddus* of numerous kinds, and *kheer*.

Bihar and Uttar Pradesh

A very popular item then as now was a mixture of broken barley and fried grams flavoured with salt and sugar, called *sattu*. Early literature from the region mentions *bara* (*vada*), *mungaura* (a *vada* made from *mung dhal*), *pakauri* (*pakoda*), *mithauri* (a *pakoda* spiced with *methi*), and *dubaki* which is described as a *pakoda*. boiled in water rather than fried in oil. Distinctive foods are thus evident in each area even from early times.

Maharashtra and North Karnataka

In the capital of Kalyana (a town in the Bidar district about 160 km west of Hyderabad) a king called Someshvara ruled about 1130 AD. He wrote a remarkable book called *Manasollasa*, which means "refreshers of the mind" and carries a list of royal pastimes and pleasures. One chapter is devoted to food and cooking. It is in this book, mentioned earlier in this chapter, that the idli is recorded by its Sanskrit name, *iddarika*. There are descriptions also of the *dosaka* (*dosa*), *vataka* (*vada*), *kshiravata* (*dahi vada*), *gharaka* (an *urad dhal vada*, now called *garige*), *vatika* (*wadian* made from fermented red dhal paste), *purika* (the small crisp *papdi*), and *parika* (the *bonda* of today). Wheat flour was used to make *purana*, a *paratha* stuffed with jaggery and gram paste, which is today called *puranpoli* in Maharashtra and *holige* in Karnataka. Various other stuffed-wheat *parathas* are described, like *vestika*, *manda* and *pahalika* (*polika*). Sweet, fried preparations made from wheat, called *kasara* and *suhali*, are mentioned, the latter being perhaps the *balushahi* which is coated with a glazed sugar layer.

As befits a king, meat dishes are described in detail. The pig was roasted whole and then carved to yield portions called *sunthakas*. The Rajput royalty of Rajasthan still make this item under the name *shula*. From pork too was made the *khanda* of *vapa* (rolled up "like a *panchanga*" or calendar), minced liver shaped into balls, and strips of bacon marinated in curds and spices. Meat was cooked in many ways: sometimes in sour fruit juices, sometimes after coating the pieces with gram flour like a *bajji*; or frequently by first roasting and then frying to give what were called *bhadritakas*, a sort of *kabab*. *Kheema* (mince) balls were called *vattimaka*, and meat patties, *purubhattaka*. *Rohu* fish was skinned, marinated in asafoetida and salt, dipped in turmeric water, and finally fried.

Rice *kanji*, soured by keeping it overnight, was seasoned the next morning and drunk as a beverage. This seems to have been a common drunk till recent times in many parts of India. Betel leaves eaten by the king had, of course, to be rather special: they were stuffed with five aromatic spices, namely cardamom, clove, nutmeg, mace and camphor.

Milk of course was used in a myriad ways. It was drunk by itself, or converted into buttermilk (this in the book is called *majjika*, as it still is today in Andhra Pradesh and Karnataka), curds or *rayata* (the modern *raita*). The popular *shrikhand* of Maharashtra was common then as well, under the name of *shikharini*; it was made by draining curds of all the water and beating it up with sugar, spices and colouring matter. *Channa* was obtained by precipitation from boiled milk, and was fashioned into a number of sweets. One was shaped like the eggs of the peacock (*mora*) and was

called *morendaka*. Rice flour shaped in ingenious ways and coated with sugar gave *kshiraprakara*. The same sweet, shaped in the form of rice grains, is now called *sitabhog* in Bengal.

Situated right in the heart of India, King Someshvara could draw upon the best of the foods of the many regions that lay to the west, south, east and north. Many of the foods that he described nearly 900 years ago have their modified counterparts in various parts of the country even to this day.

OceanofPDF.com

9 EUROPE VIEWS THE INDIAN SCENE

Though the sea route to India was opened up by Vasco da Gama in 1498, visitors from Europe had started coming to this country even two centuries before that. It was a long trip all the way by land, but curiosity about the fabled land of India, with its strange customs, its great wealth and its unusual food materials, drew them here.

Kerala kaleidoscope

St. Thomas, one of the apostles of Christ, was believed to have come to the Malabar coast about 50 years after the death of Jesus, and to have himself baptised 40 Jews living there to Christianity. Later, he was believed to have been martyred on a hillock near Madras which is still called St. Thomas' Mount. The Christian community that he founded has its bible in the Syriac language. They are called Syrian Christians and were the magnet that attracted many of the early visitors to Kerala.

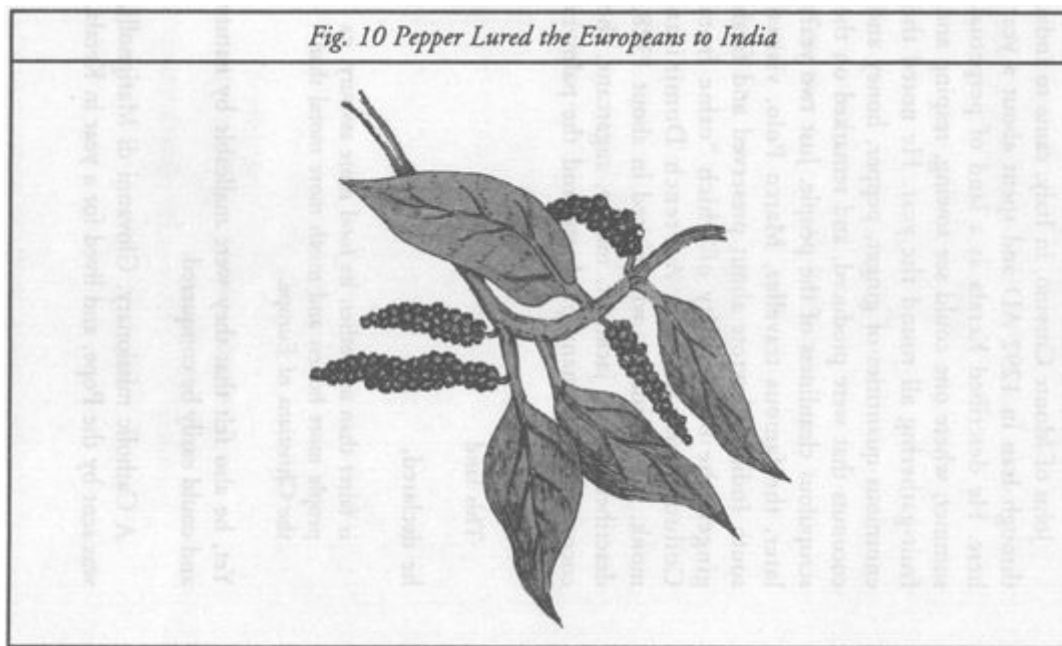
John of Monte Corvine, in Italy, came to India through Iran in 1292 AD and spent about a year here. He described Kerala as a land of perpetual summer, where one could see sowing, reaping and fruit-gathering all round the year. He noted the enormous quantities of ginger, pepper, honey and coconuts that were produced, and remarked on the scrupulous cleanliness of the people. Just two years later, the famous traveller, Marco Polo, visited south India and wrote about preserved and fresh ginger, the best quality of which "came from Coilum", now Kollam. A French Dominican monk, Friar Jordanus, who visited in about 1328, described ginger, jackfruit, mango, sugarcane, the coconut and its many products, and the palmyra palm.

This land
he declared,

is fairer than any other, its food more savoury, its people more honest and much more moral than the Christians of Europe.

Yet, he also felt that they were malleable by nature and could easily be conquered.

A Catholic missionary, Giovanni di Marignolli, was sent by the Pope, and lived for a year in Kerala.



He wrote:

On Palm Sunday AD 1357 I entered Columbum, the most famous city in the whole of India, where all the pepper in the world grows. It grows on creepers ... (in bunches) which ripen and dry on the trees ... When hardened they are struck down with staves and collected on linen cloths spread beneath.

(see Fig. 10). Ludovici di Varthema, before leaving Italy, obtained from the Pope a 10-year copyright on his writings, being well aware that others would steal and take credit for them. In Cannanore (Kannur), he noticed that vast quantities of cucumbers, melons, coconut, pepper, ginger, cardamom and other spices, mango fruit and rice were to be had. He observed that the Nair gentry of Kerala were permitted to eat venison, fish and goats, but the Brahmins could not touch animal food without losing caste. The fruits that he saw, like the sweet lime (*mosambi*) and the numerous kinds of bananas, intrigued him, but over the jackfruit he went into ecstasy, describing it as

sweet and delicious; when it is eaten it seems as though you were eating musk melons, and it appears to resemble a very ripe Persian quince. It appears also as though you were eating a preparation of honey and it also has the taste of a sweet orange.

Two visitors describe the ceremonial that attended a meal by the rulers of Kerala. Varthema noted that when the Zamorin of Calicut was to eat a meal, four priests would first offer it to the idol for some time, and after the spiritual hunger of the god had been satisfied, the meal was placed before the king as he sat on the floor. When he had finished, the left-over food was taken by the Brahmins to the courtyard and placed on the ground. On clapping their hands three times, a large number of black crows swept down and ate up the remains of the king's banquet. Duarte de Barbosa of

Portugal described the meal of a ruler of Calicut in about 1516 AD. He first chewed a betel leaf, then bathed in a pool while simultaneously worshipping, after which he donned clean clothes, and proceeding to the eating place, sat on a very low, round, wooden seat. Attendants then brought in a large silver tray (*thali*) on which were placed empty silver saucers (*katoris*). On another low stool was placed a copper pot full of cooked rice. A pile of rice was heaped on the plate, and curried meat, sauces and chutneys placed in the saucers. He ate with his right hand, using the left to pour water from a silver pitcher into his open mouth without touching it. After finishing his meal, he would return to his dais and chew on betel leaves.

Vijayanagar vista

In about 1500 AD the great capital of south India was Vijayanagar, near the present town of Bellary. Visitors were astounded at the magnificence and bounty of the place. Varthema stated that Christian visitors were warmly welcomed and could travel anywhere in absolute safety. Domingo Paes, a Portuguese merchant, lived in the city for several years, and wrote:

These dominions are very well cultivated and very fertile ... The land has plenty of rice and *jowar*, grains, beans, and other kinds of crops that are not grown in our parts; also an infinity of cotton. Of grain there is a great quantity, because, besides being used as food for men, it is also used for horses, since there is no other kind of barley; and the country has also much wheat, and that good.

Going on he says:

Wheat is not so common as the other grains, since no one eats it except the Moons (Muslims) ... The streets and markets are full of laden oxen without count ... many loads of limes come each day, such that those of Povos (in Portugal) are of no account, and also loads of sweet and sour oranges, and wild brinjals, and other garden stuff in such abundance as to stupefy one. There are many pomegranates also; grapes are sold at three bundles a fanam, and pomegranates at ten a fanam.

Elsewhere, Paes describes the birds and poultry —three kinds of partridges, quails, wild fowl, doves of two kinds, large and small, pigeons, water birds that looked like geese, poultry fowls incredibly cheap, and hares. He says that

in every street there are men who will sell you mutton, so clean and fat that it looks like pork; and you also have pigs in some streets of butchers' houses so white and clean that you could never see better in any country.

About 15 years later another Portuguese, Fernão Nuniz, was no less impressed:

Outside these cities are fields and places richly cultivated with wheat and gram and rice and millet ... and betel. The markets are always overflowing with an abundance of fruits, grapes, oranges, limes, pomegranates, jackfruit and mangoes, all very cheap.

In 1565, Vijayanagar was conquered and reduced to a ruin, now called Hampi.

Goa glimpses

As the Portuguese Catholic centre, Goa attracted a number of European visitors. Some of them were also scientists. Garcia da Orta spent as many as 35 years in Goa as a doctor tending Portuguese viceroys and officials. He developed a famous garden of his own where he grew many species of trees and medicinal plants, and wrote a book on the herbal products of India which is a mine of information of the flora of that time. He claimed that he had personally proved on himself the efficacy of drinking Ganges water. He noted that Indians keep their right thumbnail pointed so as to remove the mid-rib of the betel leaf before chewing it. Garcia da Orta mentions that the name 'coconut' is derived from the resemblance of the nut to a monkey's face, which in Spanish is called 'coco'. John Huygen von Linschoten, another botanist, thought Indian melons less sweet than those of Spain. He listed many fruit trees and described in great detail what he had observed of products manufactured from the sugarcane. Christoforas Acora, writing in 1578, catalogues among other information the various kinds of asafoetida imported into India from Afghanistan.

A number of Jesuit priests left extensive collections of letters written from India. Father Monserrate (1581-1600) spent a while in the court of the Emperor Akbar, and his observations of the food there have been noted in Chapter 7. Gonsalves Rodrigues was of the opinion that the land between Belgaum and Bijapur in northern Karnataka was

very black and fertile, and very flat ... As it is such excellent land, all food grows abundantly with only the dews (rains) ... it seems the most fertile soil imaginable.

Michael Pinheiro noted that Jain monks,

live in poverty and accept in alms only what is necessary for daily sustenance. They drink only hot water, because water has a soul that will be killed if drunk without being heated.

Pedro Texeira (1587) describes the two kinds of wine made from the palmyra palm: there was *sura* or toddy, and distilled *araca* or *arrack* into which is thrown

dry grapes, which takes off its roughness and sweetens it.

The British and others

Perhaps the first Britisher to come to India was a Jesuit, Thomas Stevens, who stayed in India for 40 years and even died in Goa. He learnt Konkani and Marathi and described his experiences in a series of letters to his father.

Coarse cinnamon (*dalchini*) grows here in this country,

he accurately observes,

but the best cinnamon comes from Ceylon and is pilled (stripped) from the young trees.

Ralph Fitch (1583) waxed eloquent that the palmyra was

the profitablest tree in the world,

because of the numerous valuable products that it yielded. Camphor he noted came to India from China, but the best types were from the island of Borneo in Indonesia. William Finch (1608-11) noted that from Agra to Lahore

the way is set on both sides with mulberry trees.

The chaplain to Sir Thomas Roe, himself the first ever British ambassador to the court of Jahangir, was the Rev. Edward Terry. His writings strike one as those of a person full of sympathy and affinity for what he saw in India:

They have not many toast or baked meats but stew most of their foods. Among many dishes of this kind, I will take notice but of one they call *deu pario* (*dopiyaza*) made of venison cut in slices, to which they put onions and herbs, some roots, with a little spice and butter: the most savoury meat I ever tasted.

At a dinner given by the nobleman, Asaf Jah, brother of Nurjahan (who inspired the Taj Mahal), Terry states a nice little mathematical puzzle:

The ambassador,

he says,

had more dishes by ten, and I less by ten than our host had, but for my part I had fifty dishes.

You can easily work that out! Terry described the wheat bread that he found in India as *panis pane melior*, in other words a super bread. He noted that the emperor Jahangir, wherever he might be travelling, drank only Ganges water which was carried to him in fine copper jars, well-tinned on the insides.

A Portuguese priest, Fray Sebastian Manrique, wrote that Agra, which for some time during Akbar's reign was the capital of the Mughal Empire, had

entire streets wholly occupied by skilled sweet-makers with dainties of all sorts,

while in Lahore, the

brilliantly lighted bazaars had a great number of occupied tents or cook-shops exhaling the aroma of spicy dishes.

Manrique spoke of three kinds of bread in north India: paper-thin *chappatis*, finger-thick bread (probably *naan*) and a sweet form called *khjuru*, covered with sugar and poppy seeds. He also observed that in Bengal neither the chicken nor its eggs were eaten, and further that both the Rajputs and Sikhs considered the meat of the wild boar a great delicacy.

Two Frenchmen now join the travellers' caravan. Jean-Baptiste Tavernier was a French jewel merchant who came to India no less than six times between 1640 and 1685, in fact even dying while on his way here once more. He noticed that the popular evening meal of the poor farmer was a *khichdi* made of rice and green gram. Francois Bernier was a doctor who ministered to the Emperor Aurangzeb. He was

astonished at the enormous variety of imported fresh fruits to be had in Delhi all the year round, and their vast consumption. On his many travels all over India he carried with him enough rice to eat for five or six days, besides sweet biscuits flavoured with anise, and lime and sugar. He also recorded that he carried always a small linen bag with an iron hook to drain out curds (*dahi*, which he spells as 'days')

nothing being considered so refreshing in this country as lemonade with days.

Bengal, he observed, was so fertile and cheap that the Europeans had a proverb among themselves which said that

The kingdom of Bengal has a hundred gates open for entrance, but not one for departure.

An Italian, Niccolao Manucci, spent the entire second-half of the 17th century in India, coming here as a young lad of only 15. It is amusing to read his reactions on first chewing a betel leaf:

My head swam to such an extent that I feared I was dying. It caused me to fall down; I lost my colour, and endured agonies.

Manucci also tells us that the mangoes of Goa were called Affonso (the Aiponso of today), Carreira and Babia. To this list we can add Pereira, later turned into Pairi. These were all produced by grafting, a technique practised by the Portuguese. Manucci also describes very accurately three kinds of jackfruit, namely the *barca*, *papa* and *pacheri*, one of which actually grows on the stem under the ground.

Robert Clive won the Battle of Plassey on 23rd June, 1757. This event laid the foundation of the British Empire in India, and Englishmen came in much larger numbers and stayed for long periods. The abundance of food was a constant theme in their letters home. As an example, here is an extract from a letter written from Calcutta in 1780 by one Mrs. Eliza Fay, the wife of a lawyer and herself a dressmaker:

We dine at 2 o'clock in the very heat of the day ... a soup, a roast fowl, curry and rice, a mutton pie, the fore-quarters of lamb, rice pudding, very good cheese, fresh churned butter, and excellent Madeira (wine).

She also notes that

eatables are very cheap.

10 THE TASTE OF THE NEW WORLD

Spain and Portugal were active seafaring nations in the 15th century. Christopher Columbus of Spain discovered America, thinking it was India, in 1492, thus opening up the New World of North and South America. Six years later, Vasco da Gama of Portugal arrived at Calicut in the real India, opening the way for subsequent colonization of the country. Since the purpose of both these expeditions was trade, the early navigators were remarkably active in looking for plants and trees in South America and Mexico that were likely to be of commercial value. They would then carry these plants to their eastern settlements and attempt to grow them there. Climatically, of course, the countries of the southern hemisphere shared a good deal in common. It is to these early pioneers that we in India, owe many of the fine food plants which we take for granted today. The tomato and the potato, the papaya and the guava, the groundnut and the cashewnut, tea and coffee all came to us in this way. And, above all, we got the chilli: you may well exclaim in surprise and rub your eyes, but it is really true that just 300 years ago, the chilli was hardly known and little used in India! Today we cannot even think of cooking in our country without using chillis.

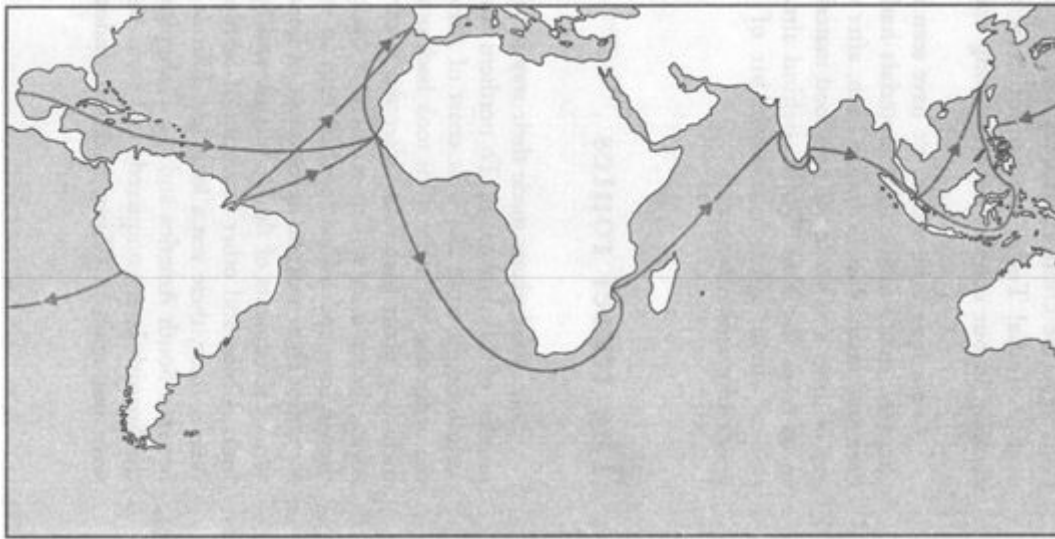
A very long time ago, as we have seen in earlier chapters, many major food materials had already been well established In India. Then, after this long gap of time, another lot of new food materials came to us from the New World. Behind almost each one of them, there is a little bit of history, geography and romance.

The travel routes

These food plants made their way to India by a number of different routes. To northern Brazil slaves were brought from the west coast of Africa across the ocean and the' same ships took bade other cargo, including plant materials. After the latter reached Africa, the sea-route to Goa on die west coast of India, though long and arduous, was by then in regular use for trade. There was also an exchange of flora between various settlements of the Portuguese and Spanish in India, China and other countries of Southeast Asia. Besides these, there was a long and difficult sea-route

between South America and the Philippines, which the Spanish later conquered, and from there ships went out again to China, Malaysia, Indonesia and India. Because of these numerous exchanges, the same food plant would sometimes reach India from more than one direction, or sometimes it would be different varieties of the same food plant (see Fig. 11). One example is tobacco; this plant is not a food, yet beedis and cigarettes made from it are "drunk" in the Indian languages! Tobacco was brought to the east coast of Andhra Pradesh, as well as to the Gujarat west coast, at different times; later, tobacco plantations flourished in both places.

Fig. 11 How American Food Plants Reached the East



Fruits

The papaya is everywhere in India, but it is actually American in origin. In fact, its name in Kannada is "foreign-tree" or *parangi-mara*, clearly showing its origin from abroad. You remember from Chapter 7 the *Ain-i-Akbari*, written in 1590, in which a list is given of the fruits offered in the Delhi market at the time: the papaya does not feature in it. However, just 8 years later, Linschoten (the botanist about whom we wrote in the earliest chapter) tells us that it came to India from the West Indies through the Philippines and Malaysia, saying also that it "much resembles a melon". Apparently, the Punjabis thought so too, and called the papaya, *kharbuza*, which is their name for the melon. The English name papaya or pawpaw seems to derive from the Peruvian "papaie". The green pieces of the plant carry an enzyme that helps tenderize meat and make it cook more easily, and this knowledge has accompanied the papaya to every country to which it has travelled.

The word guava comes from the Spanish 'guajava', which is pronounced gua-hava. It originates from South America, where it was domesticated a long time ago. By the late 17th century, it was well established in India.

Our name for the sapota is *cheeku*. This sounds really Indian, doesn't it? But you are in for a disappointment. It is really the Spanish name for this fruit, which is a product of Mexico, in Central America. At one time, the trunk of the tree used to be tapped for its latex or gum which was called "chicle;" this was the original chewing gum of the Mexicans, and later on of the United States. A well-known brand of chewing gum even today is called after this name (Chiclets). These trees grow very large, and young men called 'chicleros' were experts in clambering up to the top and tapping the gum from there. Today, the demand for chewing gum is so great that synthetic gums made in factories have completely replaced the chicle which was once used. Anyway, we still have the delicious *cheeku* fruit to enjoy.

How many of you have tasted the butter-fruit? It looks very much like a green pear, and in Spanish it is called the avocado. It has a big seed and a pale-yellow, creamy flesh with a delicate nutty flavour. Three varieties of the avocado seem to have come to us from Central America, which is its original home. The variety which came from Mexico bears rather small fruit; those from Guatemala and the West Indies carry much larger ones. The butter-fruit seems to have reached us long after the other fruits that we have just read about, but already the huge tree has run wild all over the Western Ghats of south India.

Another fruit which is not very common, but again grows wild in the Western Ghats, is the so-called passion fruit. This is a climbing plant which bears round purple or green-to-yellow fruits in enormous quantities. The jelly-like material inside is not really satisfying to eat, but makes a tasty drink. The plant is so called because the flower seems to carry the marks of the Passion of Christ's crucifixion. There are the five dark spots indicative of his wounds when nailed to the cross, and there is also a beautiful corolla suggestive of his crown of thorns.

The pineapple probably originates from Paraguay in South America. However, it quickly took root in India, and from the *Ain-i-Akbari* we know that in 1590 it was being sold in the Delhi bazaar at a price of four copper coins each (called dams). This was rather expensive, because the same money would buy ten mangoes. The name pineapple was given to it because the fruit looks like an enlarged version of the cone of the pine tree, while its taste is sweet, like an apple. A common Indian name for the pineapple is *ananas*. This was given to it by the Spanish from one of its original South American names, which was 'nana'.

Popular vegetables

The Spanish reached South America to find growing there a popular orange-coloured vegetable which was indigenous to the mountainous Andes region. One of the local names for it was 'tomatl', which gave rise to the name 'tomato' in English. The fruit was taken to Europe by 1550 and was very quickly accepted in Italy, since it made an excellent accompaniment to the cheese used with common dishes like macaroni and spaghetti. Later, the tomato also caught on in other parts of Europe. It was brought to India by the English perhaps around 1750, long after the other new plants that we are talking about. Though there are many varieties of tomato, which differ in size and shape and colour, they are all variations of a single minor species.

The potato, too, is from the Andean region, where it was called 'papa'. The sweet potato was called *batata*, and somehow when the potato was taken to Europe it got the wrong name and came to be called potato. This caused a lot of confusion later, because it is difficult to tell which one the writer is speaking about. For example, we read that the potato was served at a famous banquet given in 1615 by Asaf Jah to Sir Thomas Roe, at which the latter's chaplain, the Rev. Edward Terry, was present. However, the item served was perhaps the sweet potato. Another visitor, John Fryer, noted the potato growing in Surat and Karnataka in 1675, but again we cannot be sure which plant is meant. However, in 1780 the Governor General of India, Sir Warren Hastings, received a gift of a "basket of potatoes"; this was considered enough of an event for him to invite members of his council to dinner to share the rare treat. We can be sure that it was really potatoes this time! The British brought the potato to India, but we cannot be quite sure how long it was before the dinner of Warren Hastings that this event took place.

The large, long root of the tapioca is now a staple food in the state of Kerala. It is a plant as tall as a man, with huge green leaves, and can be seen growing everywhere in that state. It had already been domesticated as a food crop in Brazil and Venezuela. It was carried from Brazil to Africa, from where it was brought to India. The tapioca rapidly became popular in Assam, but hardly anywhere else. How did it come to be so popular in Kerala? About a hundred years ago, in 1880, there was a serious famine there which left many people starving for want of food. The ruler of Kerala at the time was Vaishakam Thirunal and he was determined that this should not happen again. He had trials carried out using several food crops to see which one would be able to ensure a steady supply of food, and the tapioca won hands down. A crop of tapioca yields about 8 000 food calories to the hectare, which is nearly four times what rice can give. Moreover, it can be easily grown in the backyard of every home, and simply left in the ground and pulled up wherever the roots are needed for food. It seems

providential that a foreign plant proved to be the salvation of people in a distant country.

OceanofPDF.com

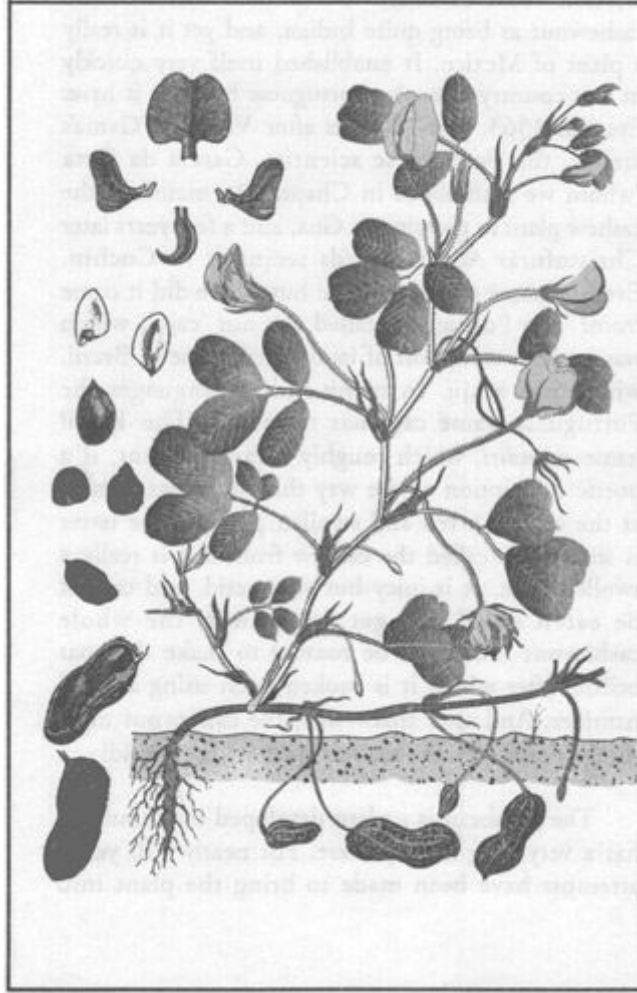
11 MORE EXOTIC DELIGHTS

We have just read of the wonderful fruits and vegetables that came to us when the New World and the Old got in touch with each other about 600 years ago. The fruits and vegetables that we have read about were only part of the story. We are beholden to the South American subcontinent for several other useful food products.

Valuable nuts

Today the groundnut is India's main oilseed crop. We also have a number of other sources of cooking oil in India, like the coconut, sesame (*til*), mustard, and safflower (*kardi*), but even so, nearly two-thirds of all our vegetable oil comes from the groundnut. Yet it is strange to find authority ¹ on the subject, just a hundred years ago, writing that "the oil is hardly known". The groundnut was developed in South America a long time ago. It has been found in ancient tombs in Peru, dated to about 900 BC. Botanists tell us that the groundnut actually originated long before that in Brazil, and that further diversification of the plant took place both there and in the neighbouring country of Peru. The Brazilian type of groundnut is an erect plant bearing large pods which have only slight constrictions. On the other hand, the Peruvian type is a runner, and the seeds are small and humpbacked, with pronounced constrictions and little shoe-strings attached to them. Both these types of nuts came to India at different times and by different routes; the Brazilian form by way of Africa, and the Peruvian through the Philippines and China. Today, the distinction between these two forms has almost vanished since they have been extensively crossed, but they still tend to be either erect or runner-like in nature. The farmer prefers the first type, since it is easier to harvest by simply pulling it out of the ground in one place. The groundnut has acquired this name because the flowers bury themselves in the soil after fertilization, and the nuts literally develop under the ground. Today, we can hardly think of the groundnut as being American, so thoroughly has it entered our *uppurna*, *chikki*, and savoury *besan* snacks, apart from our oilseed-crushing industry (see Fig. 12).

Fig. 12 The Groundnut Develops Below the Ground



Some of you may have seen the rather untidy-looking cashewnut tree growing wild along the western coast of India. We tend to think of the cashewnut as being quite Indian, and yet it is really a plant of Mexico. It established itself very quickly in this country after the Portuguese brought it here. Even in 1563, just 70 years after Vasco da Gama's arrival, the Portuguese scientist, Garcia da Orta (whom we read about in Chapter 9), mentions the cashew plant as growing in Goa, and a few years later Christoforas Acora records seeing it in Cochin. Even its name sounds Indian: but where did it come from? The Portuguese called the nut 'caju', which was itself a corruption of its original name in Brazil, which was *acaju*. In many Indian languages the Portuguese name *caju* has remained. The Tamil name *mundiri*, which roughly means in front, is a poetic description of the way the nut hangs outside at the end of a red and swollen portion; the latter is sometimes called the cashew fruit, but is really a swollen stalk. It is juicy but very acrid, and cannot be eaten at all. To get at the nut, the whole cashewnut must first be roasted to make the coat brittle, after which it is

broken open using a small hammer. And after that—well, the cashewnut must surely be one of the tastiest nuts in the world!

The soyabean is a plant developed in China and has a very long history there. For nearly 150 years, attempts have been made to bring the plant into India and cultivate it here as a *dhal*. These failed because the soyabean does not cook well into a soft product, and has also a peculiar beany flavour. Even the efforts of Mahatma Gandhi in the 1930s to popularize the soyabean did not have any effect. However, in the last twenty years new varieties of the soyabean have again been imported from America, and it is now being grown in very large quantities here. The seed is not eaten like a *dhal*. Instead the oil from it is removed in large factories, and the residual oilcake is a rich source of protein, used either to feed cattle, or to make all sorts of processed foods for us to eat and drink, like soya yoghurt, crisp snacks and textured foods like nuggets (said to resemble meat). Soya milk can be given to certain infants who are allergic to the lactose in cow and buffalo milk.

The sunflower is a common garden plant with pretty yellow flowers and a small black centre. These were developed, especially in Bulgaria and Russia, into huge plants with enormous flowers, the centre of which is a big black circle packed with seeds that are filled with oil. It is these seeds which were brought from Russia into India about thirty years ago and acclimatized to our agricultural and climatic conditions. Many varieties have now been developed, and the sunflower is grown in very large quantities in south India and Maharashtra. The oil is becoming popular as a cooking oil. The roasted seeds can be cracked and eaten, and have a pleasant nutty flavour.

Adding pleasure to eating and drinking

We had a glimpse in the last chapter that chillis are not really Indian. These wonderful materials were brought to India from Mexico, perhaps in the late 16th century. They took a little while to catch on, but in about a hundred years, the use of chillis spread to every part of India. Before that, it was pepper that was used to give the pungency that is so characteristic of Indian food. In one of the sections of the *Ain-i-Akbari*, written in 1590, there is a list of 50 dishes cooked in Akbar's court: all of them use only pepper to impart spiciness. In most Indian languages, the name for the chilli is simply a variation of the earlier name for pepper in the same language. For example, in Hindi, we say *kalimirch* (black pepper) for pepper and *harimirch* (green pepper) for chilli. In Tamil, the word for pepper is *milagu* and that for chilli is *milagai* (= *milagu-kai* (pepper+fruit)). In Kannada, the words are *karimenasu* and *menasinkayi*. Try this exercise in your own language.

It is not difficult to understand why the chilli so quickly replaced pepper in our cooking. While the pepper vine grows almost only in Kerala, chillis can be grown in almost every backyard, or cultivated in fields, all over the country. Thus, they were easily available everywhere at a low price. All the many varieties that we know come to us from Mexico and none of them was developed afterwards in India. These include the green chilli, red chilli, long red chilli, very small and very hot green bird chilli, and the large mild capsicum. To make chilli-powder, the long bright-red variety with thin skins can be dried in the sun, and ground either with its seeds to give more pungency, or without it to give a milder chilli-powder. In fact, it is no exaggeration to say that the humble chilli from Mexico really revolutionized the food of India.

On awakening, the first thing we crave in the morning is a cup of tea or coffee. So popular have these two drinks become, that one wonders what people drank in earlier days. Both these plants are not very old in the country. Tea plants had long been grown in almost a wild state in Manipur and in the adjacent hilly regions of Burma and China. This was a tall, hardy plant with thick, leathery leaves. In about 1820, the British were finding it difficult to obtain tea from China. They then decided to develop plantations in India, and after some initial doubts decided to use the hardy plants found in Assam. These were enormously successful when the leaves were first offered for sale by auction in Calcutta, and later in London. From about 1840, private plantations came up with great speed, first in Assam and later in the hilly regions of south India. Today the Assam tea plants have been hybridized with Chinese and Vietnamese varieties. The leaves that result from these crosses give us a range of colour, flavour and aroma in the tea-brew. These qualities are further developed during the drying, fermenting, sorting and blending operations by which all tea-leaves are processed before packing. We see, therefore, that tea-drinking on a large scale is barely 150 years old in India.

Coffee has a slightly longer history. The Arabs introduced it long before the arrival of the Europeans. In 1616, Edward Terry describes "cohha" and in 1700, Alexander Hamilton talks of sharing "a dish of coffee" with a certain Nawab. The seeds from which this coffee was made were imported from Arabia or Ethiopia. In the early days, coffee-drinking was confined to the countries where the seed was grown, but in the 16th and 17th centuries the habit spread all over Arabia, and from there to Turkey and later to England. The legendary coffee-shops of the 18th century became the meeting place of great literary figures like Samuel Johnson and Oliver Goldsmith. The name 'coffee' is from the Arabic word "kahwah" which originally only meant a brew. In India, there is quite a romantic story round the planting of coffee. A Muslim divine called Baba Budan went on a pilgrimage to Mecca in 1720. When he returned, he brought back with him seven green coffee berries which he grew outside the cave in

which he lived in the hills near Chikmagalur in Karnataka. These hills are now named after him, and the descendants of those original seven seeds can still be seen growing there. About a century later, the Britishers cleared the jungles on the Western Ghats of south India and started coffee plantations. These expanded rapidly, and today India is a major producer and exporter of coffee beans. The coffee berry is a beautiful red colour when it grows on the tree. It is plucked and the outer sticky pulp is first removed, after which the rest is left to ferment before complete removal. The two halves of the seed are covered with a tough parchment, which is detached by scraping it away. The green bean is then roasted until it become dark brown in colour and quite brittle. Grinding gives a dark brown powder from which our wonderfully aromatic coffee is brewed.

A couple of puzzles

We have all eaten the *sitaphal* or *sharifa* with its green knobs on the surface and delicious white pulp inside. Did it come to us from South America, as is likely, or was it already known earlier in India? A British visitor mentions the fruit in India even in 1632, and a century ago, regular forests of wild *sitaphal* were to be found in central and western India. Of course, all these descriptions occur after the arrival of the Europeans. However, there are some intriguing earlier references. The *Charaka Samhita* in Sanskrit, which is nearly 2000 years old, mentions a fruit called the *sitaphala*. Other names like *ravanamalika* and *kashmiri-amlika* occur in a book written about a thousand years ago, and these may refer to the present *sitaphal*. There is also some sculptural evidence in the Buddhist stupa at Bharhut, which was built in the 2nd century BC, where certain stone representations of trees and fruit do look rather like the *sitaphal*. The evidence is, therefore, rather conflicting whether the *sitaphal* was known here earlier, or came to us from the New World, though the latter is much more likely.

While it is true that the large-scale movement of plants from South and Central America to India occurred only after European navigators opened up the two countries, there is evidence that the New World was not totally unknown to people from India. The lotus and the elephant, both unknown in America, occur on the temple ruins of the Mayan cities. Ancient Indian astronomers also had postulated a land mass positioned where South America is.

We enjoy a roasted maize cob or *butta* with a little salt and pepper. Archaeologists and scientists have definitely proved that maize had slowly developed in Mexico over a long period of 7 000 years, from a wild grass, with a tiny little head, to a long cob. Every step in this enlargement of the corn cob has been traced. Did it come to India,

like so many other plants, with the Spanish or Portuguese, or was it here even before the discovery of the New World by Columbus?

The Indian name *makka* may only mean that the plant came to us from the direction of Mecca or the Middle East, because the word for sorghum (*jowar*) is *Durrah-i-makka*, and the maize cob does resemble the head of the *jowar*. The word *butta* may have been derived from the Pali word meaning to eat, or the Sanskrit word for a large bean. There is no reference to the maize plant in the early writings in Sanskrit, and written we begin to find references to it only after the 16th century. All this would seem to suggest that the maize had come to us, as many other plants did, from South America.

However, there are some peculiar findings. We find some very primitive forms of the maize plant grow in Assam. Even if these came from outside India, this must have been a long time ago and, in any case, long before Columbus. A piece of pottery found in Madhya Pradesh and dated 1435 AD has a strong impression of the orderly maize grains on the cob, and no other plant is likely to have yielded this impression. The temple of Somnathpura near Mysore was built in the 12th century AD. On the outside walls are a number of stone figures of goddesses holding in their left hands what look remarkably like maize cobs, some even with silken tassels.

While the historic puzzles remain, let us enjoy eating the *sitaphal*, and tackling a roasted *butta* (ear of corn)!

¹ George Watt, author of the 10-volume *Dictionary of Economic Products of India*

Bibliography

- Abul Fazl, *The Ain-i-Akbari*, trans. H. Biochmann (1871), Aadiesh Book Depot, Delhi, repr. 1965
- Achaya K.T., *Indian Food. A Historical Companion*, Oxford University Press, New Delhi, 1994
- Achaya K.T., *A Historical Dictionary of Indian Food*, Oxford University Press, New Delhi, 1998
- Jeannine Auboyer, *Daily Life in Ancient India*, Indian Council of Agricultural Research, New Delhi, 1964
- Ghurye G.S., *Vedic India*, Popular Prakashan, Bombay, 1979
- Ibn Battuta, *Travels in Asia and Africa-1325-1354*, trans. H.A.R. Gibbs, Routledge and Kegan Paul Ltd., London, 4th impr., 1957
- Kangle R.P., *The Kautilya Arthasastra*, Motilal Banarsidass, Delhi, 2nd ed., 1972, repr. 1986
- Khare R.S., *The Hindu Hearth and Home*, Vikas Publishing House, India, 1976
- Khare R.S., *Culture and Reality*, Indian Institute of Advanced Studies, Shimla, 1976
- Jois S.N.Krishna (ed.), *Supa Sastra* of Mangarasa (AD 1516), University of Mysore, 1969
- Sastri KA Nilakanta, *History and Culture of the Tamils*, Firma Mukhopadhyay K.L., Calcutta, 1964
- Om Prakash, *Food and Drink in Ancient India*, Munshiram Manoharlal, Delhi, 1961
- Possehl G.L. (ed.), *Food Economy of the Harappans*, Oxford and IBH Publishing House, Delhi, 1982
- Ray Priyadarajan and Gupta Hirendra Nath, *Caraka Samhita: A Scientific Synopsis*, Indian National Science Academy, New Delhi, 1980
- Ray Priyadarajan, Gupta Hirendra Nath and Roy M., *Susruta Samhita: A Scientific Synopsis*, Indian National Science Academy, New Delhi, 1980
- Sahu K.P. *Some Aspects of North Indian Social Life 1000-1526 AD*, Punthi Pustak, Calcutta, 1973
- Shrigondekar K.L., *Manasollasa of King Someshwara*, Gaekwad's Oriental Series, Baroda, 1939
- Simmonds N.W. (ed.), *Evolution of Food Plants*, Longman, London and New York, 1976
- Iyengar M. Srinivasa, *Tamil Studies* (1913), Asian Educational Services, New Delhi, repr. 1982
- Iyengar P.T. Srinivasa, *History of the Tamils* (c. 1932), Asian Educational Services, New Delhi, repr. 1983
- Vogel Claus, *Vaghhata's stanghrdayasamhita*, Kommissionsverlag Franz Sterner Gmbh, Wiesbaden, 1965

GLOSSARY OF INDIAN AND NON-ENGLISH WORDS

acaju

ad

adai

Ain-i-Akbari

Amir

Amla

amra

ananas

appam

areca

aral

Aranyakas

arisi

Arthasastra

Atharvaveda

Ayurveda

bhadra

bhadritaka

bhatura

Bhisma

birinj

biriyani

Bismillah

bonda

boondhi

brahman, brahmin

Brahmanas

Brahmi

brinjal

butta

caju

chana

Chandi Mangala

<i>badi</i>	<i>Chappati</i>
<i>bajji</i>	
<i>bajra</i>	<i>Charaka Samhita</i>
<i>balushahi</i>	<i>chaval</i>
<i>barca</i>	<i>cheeku</i>
<i>batata</i>	<i>chekku</i>
<i>ber</i>	<i>channa</i>
<i>besan</i>	<i>chicle</i>
<i>chicleros</i>	<i>fanam</i>
<i>chikki</i>	<i>fuqqa</i>
<i>chinchapala</i>	<i>Ganga</i>
<i>chitranna</i>	<i>garige</i>
<i>chivda,</i>	<i>ghani</i>
<i>cholam</i>	<i>ghee</i>
<i>chom-la</i>	<i>gingelli</i>
<i>chula</i>	<i>guda</i>
<i>coco</i>	<i>gulab-jamun</i>
<i>cohha</i>	<i>gur</i>
<i>copra</i>	<i>guvaka</i>
<i>dadhanwat</i>	
<i>dadhi</i>	<i>haldi</i>

dahi

haleem

dahi-wada

halwa

dalchini

harimirch

dam

harisa

days

hilsa

deu pario

holige

dhal

huen-lo

dhokla

hujjab

dopiyaza

bhadra

dosa(i)

iddarika

dosaka

idi-appam

Draupadi

idli

Dubaki

i-gis

dumpukt

illisa

Durrah-i-makka

injivera

ennai

faluda

jambhira

jeera

jamun

jom

kheer

jowar

khichdi

	<i>khiri</i>
<i>kabab</i>	<i>khisti</i>
<i>kaccha</i>	<i>khjuru</i>
<i>kahwah</i>	Kshatriya
<i>kali-mirch</i>	<i>kshiraprakara</i>
<i>kangni</i>	<i>kshiravata</i>
<i>kanji</i>	<i>kshirika</i>
<i>kapittha</i>	<i>kung-ja-tin-mai</i>
<i>karambo</i>	<i>kur</i>
<i>kardi</i>	
<i>karela</i>	<i>laddu</i>
<i>kari ,</i>	<i>lal-mohan</i>
<i>karimenasu</i>	<i>lazizan</i>
<i>karipattha</i>	
<i>kasara</i>	
<i>Kashmiri-amlika</i>	<i>ma</i>
<i>kashkh</i>	<i>maanga</i>
<i>katori</i>	<i>madhu</i>
<i>Kauravas</i>	<i>madhuka</i>
<i>kedli</i>	<i>Mahabharata</i>
<i>keerai</i>	<i>mahua</i>

kesar

maida

Khaja

majjika

Khanda

makka

khandvi

mamsaudana

khaman

Manasollasa

kharbuza

manda

khatiba

masoor

kheema

masha

masura

mashaikh

matki

nimbu

mattar

nisa

menasinkayi

Nisada

methi

niyor

milagai

numbaka

milagu

oma

milagu-kayi

pacheri

mithauri

pahalika

mocha

pakoda

modaka

palak

mora

palao

morendaka

palas

<i>morram</i>	<i>panasa</i>
<i>mosambi</i>	<i>panchanga</i>
<i>mudga</i>	<i>Pandavas</i>
<i>mundiri</i>	<i>paneer</i>
<i>mullai</i>	<i>panis pane melior</i>
<i>mung</i>	<i>papa</i>
<i>mungaura</i>	<i>papad</i>
<i>murukku</i>	<i>papaie</i>
<i>Mysore pak</i>	<i>papdi</i>
<i>naan</i>	<i>parangi-mara</i>
<i>nadu</i>	<i>paratha</i>
<i>nai</i>	<i>parika</i>
<i>nana</i>	<i>parpata</i>
<i>narikela</i>	<i>payesh</i>
<i>nariyal</i>	<i>Perumpunnaru</i>
<i>neem</i>	<i>pheni</i>
<i>nellikai</i>	<i>pitha</i>
<i>nimbaka</i>	<i>saiyid</i>
<i>polika</i>	<i>Samaveda</i>
<i>pori-kari</i>	<i>samosa</i>
<i>pucca</i>	<i>sandesh</i>

<i>puja</i>	<i>santhra</i>
<i>puli</i>	<i>sarson</i>
<i>pulingari</i>	<i>sarson-ka-saag</i>
<i>punjistha</i>	<i>sattu</i>
<i>Purana</i>	<i>sattwik</i>
<i>puran-poli</i>	<i>seekh kabab</i>
<i>puri</i>	<i>se-gis-i</i>
<i>purika</i>	<i>semiya</i>
<i>purubhattaka</i>	<i>shakkara</i>
	<i>Shaman</i>
<i>qazi</i>	<i>shammi</i>
<i>qaliya</i>	<i>Shantanu</i>
<i>qutab</i>	<i>sharifa</i>
	<i>shikampoor</i>
<i>rabadi</i>	<i>shikharini</i>
<i>ragi</i>	<i>Shiva</i>
<i>rai</i>	<i>shorfa</i>
<i>raita</i>	<i>shrikhand</i>
<i>Ramayana</i>	<i>shukto</i>
<i>ras-malai</i>	<i>shula</i>
<i>rasogolla</i>	<i>shulla</i>

ravanamlika

shurba

rayata

shurbdariya

Rehala

shurpa

Rigveda

sigri

rohituka

simat

rohu

tuvarai

rui

tuvarika

sitabhog

Upanishads

sitaphal

uppuma

sitaphala

urad

Si-yu-ki

sramana

vada

subunia

Vaisya

sufiyana

vapa

Sudra

varagu

suhali

varisi

sukhadi

vataka

sunthaka

vatika

sura

vattimaka

Susrutha Samhita

Veda

svanin

vedhami

	<i>vennai</i>
<i>tael</i>	<i>vestika</i>
<i>taila</i>	Vishnu
<i>tallita-kari</i>	<i>vrihi</i>
<i>tamasic</i>	<i>vrntaka</i>
<i>tandoori</i>	
<i>tenga</i>	
<i>thali</i>	<i>wadian</i>
<i>thambula</i>	
<i>thennai l</i>	<i>yava</i>
<i>til</i>	<i>yajna</i>
<i>tila</i>	<i>Yajurveda</i>
<i>toor</i>	
<i>tomatl</i>	<i>zafran</i>
<i>toria</i>	Zamorin
<i>tun</i>	<i>zebu</i>
<i>tuvar</i>	<i>tuvarai</i>